

AP EAMCET-2017-Engineering-26 Apr-Shift-1
Mathematics

1. Let $A = \{-4, -2, -1, 0, 3, 5\}$ and $f : A \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 3x-1 & \text{for } x > 3 \\ x^2+1 & \text{for } -3 \leq x \leq 3 \\ 2x-3 & \text{for } x < -3 \end{cases}$$

Then the range of f is

- (1) $\{-11, -7, 2, 1, 8, 14\}$
(2) $\{-11, 5, 2, 1, 8, 14\}$
(3) $\{-11, 5, 2, 1, 10, 14\}$
(4) $\{-11, -7, -5, 1, 10, 14\}$
2. Let $\mathbb{N}$ be the set of all natural number, $\mathbb{Z}$ be the set of all integers and $\sigma : \mathbb{N} \rightarrow \mathbb{Z}$ be defined by

$$\sigma(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ -\frac{(n-1)}{2}, & \text{if } n \text{ is odd} \end{cases}$$

Then

- (1) σ is one-one but not onto
- (2) σ is onto but not one-one
- (3) σ is one-one and onto.
- (4) σ is neither one-one nor onto.

3. The sum of first n terms of $\frac{1}{2 \cdot 5} + \frac{1}{5 \cdot 8} + \frac{1}{8 \cdot 11} + \dots$ is

(1) $\frac{3n}{2(3n+2)}$

(2) $\frac{3n}{3n+2}$

(3) $\frac{n}{3n+2}$

(4) $\frac{n}{2(3n+2)}$

4. If a, b, c are non zero real numbers and if the equations $(a-1)x = y+z$, $(b-1)y = z+x$ and $(c-1)z = x+y$ have a non trivial solution then $ab+bc+ca =$,

- (1) $a+b+c$
- (2) abc

(3) 0

(4) $a^2b^2c^2$

5. If a system of three linear equations in three unknowns, which is the matrix equation form of $AX = D$ is consistent, then

$\frac{\text{rank of } A}{\text{rank of } AD}$ is

- (1) Greater than or equal to one.
- (2) One
- (3) Greater than one.
- (4) Less than one.

6. If $\Delta = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ -1 & -\cos \theta & 1 \end{vmatrix}$, then Δ lies in the interval

- (1) $(2, 4)$
- (2) $[2, 4]$
- (3) $[-1, 1]$
- (4) $[1, 4]$

7. If z is complex number with $|z| \geq 5$, then the least value of

$$\left| z + \frac{2}{z} \right| \text{ is}$$

(1) $\frac{23}{5}$

(2) $\frac{24}{5}$

(3) $\frac{26}{5}$

(4) $\frac{29}{5}$

8. If $\log_{\frac{1}{\sqrt{3}}} \left\{ \frac{|z|^2 - |z| + 1}{2 + |z|} \right\} > -2$, then z lies inside

(1) a square

(2) a triangle

(3) a circle

(4) an ellipse

9. If ω is a complex cube root of unity, then for any

$$n > 1, \sum_{r=1}^{n-1} r(r+1-\omega)(r+1-\omega^2) =$$

(1) $\frac{n^2(n+1)^2}{4}$

(2) $\frac{n(n-1)}{4}(n^2+3n+4)$

(3) $\frac{n(n+1)(2n+1)}{4}$

(4) $\frac{n(n+1)(2n+1)}{6}$

10. If α is a non real root of $x^7 = 1$, then $\alpha(1+\alpha)(1+\alpha^2+\alpha^4) =$

(1) 2

(2) -1

(3) 1

(4) -2

11. $\frac{x-1}{3x+4} - \frac{x-3}{3x-2} < 0$ holds, for all x in the interval,

$$(1) \left(-\infty, \frac{-5}{4} \right)$$

$$(2) \left(\frac{-4}{3}, \frac{2}{3} \right)$$

$$(3) \left(\frac{3}{4}, \infty \right)$$

$$(4) \left(-\infty, \frac{-5}{4} \right) \cup \left(\frac{3}{4}, \infty \right)$$

12. The harmonic mean of two number is $-\frac{8}{5}$ and their geometric mean is 2. The quadratic equation whose roots are twice those numbers is,

$$(1) x^2 + 10x + 16 = 0$$

$$(2) x^2 - 10x + 16 = 0$$

$$(3) x^2 + 5x + 4 = 0$$

$$(4) x^2 - 5x + 4 = 0$$

13. If the roots of the equation $x^3 - 7x^2 + 14x - 8 = 0$ are in a geometric progression, then the difference between the largest and the smallest roots is

(1) $\frac{1}{2}$

(2) 2

(3) 3

(4) 4

14. If α, β, γ are the roots of $x^3 + px^2 + qx + r = 0$, then the value of

$(1 + \alpha^2)(1 + \beta^2)(1 + \gamma^2)$ is

(1) $(r + p)^2 + (q + 1)^2$

(2) $(r - p)^2 + (q - 1)^2$

(3) $(1 + p)^2 + (1 + q)^2$

(4) $(r - p)^2 + (r - q)^2$

15. The number of different ways of preparing a garland using 6 distinct white roses and 5 distinct red roses such that no two red roses come together is

(1) 151200

(2) 21600

(3) 43200

(4) 86400

16. There are 10 intermediate stations on a railway line between two particular stations. The number of ways that a train can be made to stop at 3 of these intermediate stations so that no two of these halting stations are consecutive, is

- (1) 56
- (2) 126
- (3) 20
- (4) 120

17. ${}^{37}C_4 + \sum_{r=1}^5 {}^{42-r}C_r =$

- (1) ${}^{38}C_4$
- (2) ${}^{39}C_4$
- (3) ${}^{41}C_4$
- (4) ${}^{42}C_4$

18. The coefficient of x^5 in the expansion of $(1+x)^{21} + (1+x)^{22} + \cdots + (1+x)^{30}$ is

(1) ${}^{51}C_5$

(2) 9C_5

(3) ${}^{31}C_6 - {}^{21}C_6$

(4) ${}^{30}C_5 + {}^{20}C_5$

19. If $x = \frac{1 \cdot 3}{3 \cdot 6} + \frac{1 \cdot 3 \cdot 5}{3 \cdot 6 \cdot 9} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{3 \cdot 6 \cdot 9 \cdot 12} + \dots$ to infinite terms, then

$9x^2 + 24x =$

(1) 11

(2) 21

(3) 31

(4) 41

20. If $\frac{5x^2 + 2}{x^3 + x} = \frac{A_1}{x} + \frac{A_2x + A_3}{x^2 + 1}$, then $(A_1, A_2, A_3) =$

(1) (0, 2, 3)

(2) (2, 0, 3)

(3) (3, 0, 2)

(4) (2, 3, 0)

21. If $\tan \alpha$ and $\tan \beta$ are the roots of the equation $x^2 + px + q = 0$,

then the value of

$$\sin^2(\alpha + \beta) + p \cos(\alpha + \beta) \sin(\alpha + \beta) + q \cos^2(\alpha + \beta) \text{ is}$$

(1) $p + q$

(2) p

(3) q

(4) $\frac{p}{p + q}$

22. $1 + \cos 10^\circ + \cos 20^\circ + \cos 30^\circ =$

(1) $4 \cos 5^\circ \cos 10^\circ \cos 15^\circ$

(2) $4 \cos 10^\circ \cos 20^\circ \cos 30^\circ$

(3) $4 \sin 5^\circ \sin 10^\circ \sin 15^\circ$

(4) $4 \sin 10^\circ \sin 20^\circ \sin 30^\circ$

23. The number of solution of the trigonometric equations

$$1 + \cos x \cdot \cos 5x = \sin^2 x \text{ in } [0, 2\pi] \text{ is}$$

(1) 6

(2) 8

(3) 10

(4) 12

24. If $\cot(\cos^{-1} x) = \sec \left\{ \tan^{-1} \left(\frac{a}{\sqrt{b^2 - a^2}} \right) \right\}$; $b > a$, then $x =$

(1) $\frac{b}{\sqrt{2b^2 - a^2}}$

(2) $\frac{a}{\sqrt{2b^2 - a^2}}$

(3) $\frac{\sqrt{b^2 - a^2}}{a}$

(4) $\frac{\sqrt{b^2 - a^2}}{ab}$

25. $\operatorname{sech}^{-1} \left(\frac{1}{\sqrt{2}} \right) + \operatorname{cosech}^{-1}(-1) =$

(1) $\sqrt{2} - 1$

(2) $\sqrt{2} + 1$

(3) $\sqrt{2}$

(4) 0

26. In ΔABC if θ is any angle then $b \cos(C + \theta) + c \cos(B - \theta) =$

- (1) $a \cos \theta$
- (2) $a \sin \theta$
- (3) $a \tan \theta$
- (4) $a \cot \theta$

27. In ΔABC , if $b \cos \theta = c - a$, (where θ is an acute angle), then

$$(c - a) \tan \theta =$$

- (1) $2\sqrt{ca} \cos \frac{B}{2}$
- (2) $2\sqrt{ca} \sin \frac{B}{2}$
- (3) $2ca \sin \frac{B}{2}$
- (4) $2ca \cos \frac{B}{2}$

28. If α, β, γ are the lengths of the tangents from the vertices of a triangle to its in-circle, then

$$(1) \alpha + \beta + \gamma = \frac{1}{r^2}(\alpha\beta\gamma)$$

$$(2) \alpha + \beta + \gamma = \frac{1}{r}(\alpha\beta\gamma)$$

$$(3) \alpha^2 + \beta^2 + \gamma^2 = \frac{2}{r}(\alpha\beta\gamma)$$

$$(4) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = r(\alpha\beta\gamma)$$

29. The triad (x, y, z) of real numbers such that

$$(3\bar{i} - \bar{j} + 2\bar{k}) = \begin{bmatrix} (2\bar{i} + 2\bar{j} - \bar{k})x + (\bar{i} - 2\bar{j} + 2\bar{k})y \\ + (-2\bar{i} + \bar{j} - 2\bar{k})z \end{bmatrix}, \text{ is}$$

$$(1) (2, -5, 3)$$

$$(2) (2, 5, -3)$$

$$(3) (-2, 5, 3)$$

$$(4) (2, 5, 3)$$

30. In $\triangle ABC$, L, M, N are points on BC, CA, AB respectively, dividing them in the ratio $1:2, 2:3, 3:5$. If the point K divides

AB in the ratio $5:3$, then $\left| \frac{\overline{AL} + \overline{BM} + \overline{CN}}{\overline{CK}} \right| =$

(1) $\frac{1}{15}$

(2) $\frac{2}{5}$

(3) $\frac{5}{8}$

(4) $\frac{3}{5}$

31. If a non zero vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\vec{j} - \vec{k}, 3\vec{j} - 2\vec{k}$ and the plane determined by the vectors $2\vec{i} + 3\vec{j}, \vec{i} - 3\vec{j}$, then the angle between $\vec{a}$ and $\vec{i} + \vec{j} + \vec{k}$ is

(1) $\cos^{-1}\left(\pm \frac{1}{\sqrt{3}}\right)$

(2) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

$$(3) \tan^{-1} \sqrt{3}$$

$$(4) \cos^{-1} \left(\pm \frac{2}{\sqrt{3}} \right)$$

32. If $\vec{a}, \vec{b}, \vec{c}$ are non-zero vectors such that $\vec{a}$ and $\vec{b}$ are not perpendicular to each other, then the vector $\vec{r}$ which is perpendicular to $\vec{a}$ and satisfying $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ is

$$(1) \frac{(\vec{b} \times \vec{c}) \times \vec{a}}{\vec{a} \cdot \vec{b}}$$

$$(2) \frac{\vec{b} \times (\vec{a} \times \vec{c})}{\vec{b} \cdot \vec{c}}$$

$$(3) \frac{(\vec{c} \times \vec{b}) \times \vec{a}}{\vec{a} \cdot \vec{c}}$$

$$(4) \frac{(\vec{a} \times \vec{b}) \times \vec{c}}{\vec{c} \cdot \vec{a}}$$

33. If the volume of the tetrahedron formed by the coterminal edges $\vec{a}, \vec{b}$ and $\vec{c}$ is 4, then the volume of the parallelepiped formed by the coterminal edges $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}$ and $\vec{c} \times \vec{a}$ is

$$(1) 144$$

- (2) 16
- (3) 48
- (4) 576

34. If the points having the position $3\vec{i} - 2\vec{j} - \vec{k}$, $2\vec{i} + 3\vec{j} - 4\vec{k}$, $-\vec{i} + \vec{j} + 2\vec{k}$ and $4\vec{i} + 5\vec{j} + \lambda\vec{k}$ are coplanar then $\lambda =$

- (1) -8
- (2) 8
- (3) $\frac{146}{17}$
- (4) $-\frac{146}{17}$

35. The variance of the following data is

x_i	6	10	14	18	24	28	30
f_i	2	4	7	12	8	4	3

- (1) 43.4
- (2) 34.3
- (3) 44.3
- (4) 33.4

36. If $\alpha \neq 0$ and the mean deviation of the observation $\{ka\}$, for $k = 1, 2, \dots, 50$ about its median is 50, then $|\alpha| =$

- (1) 2
- (2) 3
- (3) 4
- (4) 5

37. If three numbers are drawn at random successively without replacement from a set $S = \{1, 2, \dots, 10\}$, then the probability that the minimum of the chosen number is 3 or their maximum is 7, is

- (1) $\frac{1}{40}$
- (2) $\frac{3}{40}$
- (3) $\frac{5}{40}$
- (4) $\frac{11}{40}$

38. A bag P contains 5 white marbles and 3 black marbles. Four marbles are drawn at random from P and are put in an empty bag Q . If a marble drawn at random from Q is found to be black, then the probability that all the three black marbles in P are transferred to Q is

- (1) $\frac{1}{8}$
- (2) $\frac{7}{8}$
- (3) $\frac{6}{7}$
- (4) $\frac{1}{7}$

39. Given below is the distribution of a random variable X

$X = x$	1	2	3	4
$P(X = x)$	λ	2λ	3λ	4λ

$\alpha = P(X < 3)$ and $\beta = P(X > 2)$, then $\alpha : \beta =$

- (1) 2:5
- (2) 3:4
- (3) 4:5

(4) 3:7

40. If the mean and variance of the binomial variate X are $\frac{4}{3}, \frac{8}{9}$

respectively, then $P(X = 2) =$

(1) $\frac{4}{27}$

(2) $\frac{8}{81}$

(3) $\frac{16}{81}$

(4) $\frac{8}{27}$

41. $p, x_1, x_2, \dots, x_n$ and $q, y_1, y_2, \dots, y_n$ are two arithmetic progression with common differences a and b respectively. If α and β are the arithmetic means of $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ respectively, then the locus of $P(\alpha, \beta)$ is

(1) $\alpha(x - p) = \beta(y - q)$

(2) $p(x - \alpha) = q(y - \beta)$

(3) $a(x - p) = b(y - q)$

$$(4) \ b(x - p) = a(y - q)$$

42. The point to which the origin is to be shifted to remove the first degree terms from the equation

$$2x^2 + 4xy - 6y^2 + 2x + 8y + 1 = 0 \text{ is}$$

(1) $\left(\frac{7}{8}, \frac{3}{8}\right)$

(2) $\left(\frac{7}{8}, -\frac{3}{8}\right)$

(3) $\left(-\frac{7}{8}, \frac{3}{8}\right)$

(4) $\left(-\frac{7}{8}, -\frac{3}{8}\right)$

43. The incentre of the triangle formed by the straight lines

$$y = \sqrt{3}x, y = -\sqrt{3}x \text{ and } y = 3 \text{ is}$$

(1) (1, 2)

(2) (0, 2)

(3) (2, 1)

(4) (2, 0)

44. The set of all values of 'a' such that both the points $(1, 2)$ and $(3, 4)$ lie on the same side of the line $2x - 5y + a = 0$

(1) $\{x \in \mathbb{R} : x > 11\}$

(2) $\{x \in \mathbb{R} : x < 7\}$

(3) $\{x \in \mathbb{R} : x > 11\} \cup \{x \in \mathbb{R} : x < 7\}$

(4) ϕ

45. The points on the straight line $3x - 4y + 1 = 0$ which are at a distance of 5 units from the point $(3, 2)$ are

(1) $\left(1, \frac{1}{2}\right), \left(2, \frac{5}{4}\right)$

(2) $\left(-2, -\frac{7}{4}\right), \left(-3, \frac{-5}{2}\right)$

(3) $\left(4, \frac{11}{4}\right), (-1, -1)$

(4) $(7, 5), (-1, -1)$

46. The equation of the pair of lines through the point $(2, 1)$ and perpendicular to the pair of lines $4xy + 6y + 3 = 0$

(1) $xy + x + 2y - 6 = 0$

(2) $xy - x + 2y - 2 = 0$

(3) $xy + x - 2y - 2 = 0$

(4) $xy - x - 2y + 2 = 0$

47. If the pair of lines $x^2 - 16pxy - y^2 = 0$ and $x^2 - 16qxy - y^2 = 0$ are such that each pair bisects the angle between the other pair, then $pq =$

(1) $\frac{1}{8}$

(2) $\frac{-1}{8}$

(3) $\frac{1}{64}$

(4) $\frac{-1}{64}$

48. The figure formed by the pairs of lines $6x^2 + 13xy + 6y^2 = 0$ and $6x^2 + 13xy + 6y^2 + 10x + 10y + 4 = 0$, is a

- (1) Square
- (2) Rhombus
- (3) Rectangle
- (4) Parallelogram

49. A circle having centre at the origin passes through the three vertices of an equilateral triangle, the length of its median being 9 units. Then the equation of circle is

- (1) $x^2 + y^2 = 18$
- (2) $x^2 + y^2 = 81$
- (3) $x^2 + y^2 = 36$
- (4) $x^2 + y^2 = 9$

50. If a circle with radius 2.5 units, passes through the points $(2, 3)$ and $(5, 7)$, then its centre is

- (1) $(7, 10)$
- (2) $(1.5, 2)$

(3) $(3.5, 5)$

(4) $(3, 4)$

51. If $2kx + 3y - 1 = 0$, $2x + y + 5 = 0$ are conjugate lines with respect to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$, then $k =$

(1) 1

(2) 2

(3) 3

(4) 4

52. If the point of intersection of the tangents drawn at the points where the line $5x + y + 1 = 0$ cuts the circle $x^2 + y^2 - 2x - 6y - 8 = 0$ is (a, b) , then $5a + b =$

(1) 3

(2) 4

(3) -1

(4) -44

53. The angle between the two circles, each passing through the centre of the other, is

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{2}$

(3) $\frac{2\pi}{3}$

(4) π

54. The equation of the circle whose diameter is the common chord of the circles $x^2 + y^2 + 2x + 2y + 1 = 0$ and $x^2 + y^2 + 4x + 6y + 4 = 0$ is

(1) $2x^2 + 2y^2 - 2x + 4y + 1 = 0$

(2) $x^2 + y^2 - x + 2y + 4 = 0$

(3) $3x^2 + 3y^2 - 3x + 6y - 8 = 0$

(4) $10x^2 + 10y^2 + 14x + 8y + 1 = 0$

55. The equation of the parabola whose axis is parallel to the X-axis and which passes through the points $(-2, 1), (1, 2), (-1, 3)$ is

(1) $25x^2 - 2x - 65y + 36 = 0$

(2) $15y^2 + 12x - 11y + 20 = 0$

(3) $5y^2 + 2x - 21y + 20 = 0$

(4) $18y^2 - 12x - 21y - 21 = 0$

56. The tangent to the parabola $y^2 = 4ax$ from an external point P make angles θ_1 and θ_2 with the axis of parabola such that $\tan \theta_1 + \tan \theta_2 = b$. Then P lies on

(1) $y = bx$

(2) $y = x + b$

(3) $y = \frac{x}{b}$

(4) $y + x = b$

57. The line $y = 2x + \sqrt{76}$ and $2y + x = 8$ touch the ellipse $\frac{x^2}{16} + \frac{y^2}{12} = 1$. If the point of intersection of these two lines lie on a circle, whose centre coincides with the centre of that ellipse, then the equation of that circle is

(1) $x^2 + y^2 = 16$

(2) $x^2 + y^2 = 12$

(3) $x^2 + y^2 = 28$

(4) $x^2 + y^2 = (4 + \sqrt{8})^2$

58. The angle between the tangent drawn from the point $(1, 2)$ to the ellipse $3x^2 + 2y^2 = 5$ is

(1) $\frac{\pi}{2}$

(2) $\tan^{-1}\left(\frac{12\sqrt{5}}{5}\right)$

(3) $\tan^{-1}\left(\frac{12\sqrt{5}}{13}\right)$

(4) $\frac{\pi}{4}$

59. If $lx + my = 1$ is a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then

$a^2m^2 - b^2l^2 =$

(1) $l^2m^2(a^2 + b^2)$

(2) $(l^2 + m^2)(a^2 + b^2)^2$

$$(3) \frac{l^2}{m^2}(a^2 + b^2)^2$$

$$(4) \frac{m^2}{l^2}(a^2 + b^2)^2$$

60. The circumcentre of the triangle formed by the points $(1, 2, 3)$, $(3, -1, 5)$, $(4, 0, -3)$ is

(1) $(2, 2, 2)$

(2) $(1, 1, 1)$

(3) $(3, 3, 3)$

(4) $\left(\frac{7}{2}, \frac{-1}{2}, 1\right)$

61. ΔABC is formed by $A(1, 8, 4)$, $B(0, -11, 4)$ and $C(2, -3, 1)$, If D is the foot of the perpendicular from A to BC , then the coordinates of D are

(1) $(4, 5, -2)$

(2) $(4, -5, 2)$

(3) $(-4, 5, 2)$

(4) $(4, -5, -2)$

62. The equation of the parabola through $(4, 4, 0)$ and perpendicular to the plane $2x + y + 2z + 3 = 0$ and $3x + 3y + 2z - 8 = 0$ is

(1) $4x + 3y + 3z = 28$

(2) $4x - 2y - 3z = 8$

(3) $4x + 2y + 3z = 24$

(4) $4x + 2y - 3z = 24$

63. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x} =$

(1) 1

(2) 2

(3) $\frac{-1}{4}$

(4) $\frac{1}{2}$

64. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} x-1 & , \text{ for } x \leq 1 \\ 2-x^2 & , \text{ for } 1 < x \leq 3 \\ x-10 & , \text{ for } 3 < x < 5 \\ 2x & , \text{ for } x \geq 5 \end{cases}$$

then the set of points of discontinuity of f is

- (1) $\mathbb{R} - \{1, 3, 5\}$
- (2) $\{1, 3, 5\}$
- (3) $\mathbb{R} - \{1, 5\}$
- (4) $\{1, 5\}$

65. If $y = \tan^{-1} x \left\{ \frac{x}{1 + \sqrt{1-x^2}} \right\} + \sin \left\{ 2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right\}$ then $\frac{dy}{dx} =$

- (1) $\frac{2-x}{2\sqrt{1-x^2}}$
- (2) $\frac{1-2x}{x\sqrt{1-x^2}}$
- (3) $\frac{2x+1}{x\sqrt{1-x}}$
- (4) $\frac{1-2x}{2\sqrt{1-x^2}}$

66. For $x^2 - 4 \neq 0$ then the value of $\frac{d}{dx} \left[\log \left\{ e^x \left(\frac{x-2}{x+2} \right)^{\frac{3}{4}} \right\} \right]$ at $x = 3$

is

(1) 1

(2) $\frac{8}{5}$

(3) 2

(4) $\frac{8e^3}{5}$

67. If $y = \frac{\sinh^{-1} x}{\sqrt{1+x^2}}$ then $(1+x^2)y_2 + 3xy_1 + y =$

(1) 0

(2) -1

(3) 1

(4) 2

68. An angle between the curves $x^2 = 3y$ and $x^2 + y^2 = 4$ is

(1) $\frac{\pi}{3}$

(2) $\tan^{-1} \frac{2}{\sqrt{3}}$

(3) $\tan^{-1} \frac{5}{\sqrt{3}}$

(4) $\tan^{-1} \sqrt{\frac{5}{3}}$

69. A container is in the shape of inverted cone. Its height is 6 m and radius is 4 m at the top. If it is filled with water at the rate of $3 \text{ m}^3/\text{min}$, then the rate of change of height of water (in m/min) when the water level is 3 m, is

(1) $\frac{3}{4\pi}$

(2) $\frac{2}{9\pi}$

(3) 16π

(4) 2π

70. If a cylindrical vessel of given volume V with no lid on the top is to be made from a sheet of metal, then the radius (r) and height (h) of the vessel so that the metal sheet used is minimum, is

(1) $r = \sqrt[3]{\frac{V}{\pi}}, h = \sqrt[3]{\frac{V}{\pi}}$

(2) $r = \sqrt{\frac{V}{\pi}}, h = \sqrt{\frac{V}{\pi}}$

(3) $r = \sqrt{\pi V}, h = \sqrt{\pi V}$

(4) $r = \sqrt[3]{\frac{\pi}{V}}, h = \sqrt[3]{\frac{\pi}{V}}$

71. For the function $f(x) = (x-1)(x-2)$ defined $\left[0, \frac{1}{2}\right]$, the value of 'c' satisfying Lagrange's mean value theorem is

(1) $\frac{1}{3}$

(2) $\frac{1}{5}$

(3) $\frac{1}{7}$

(4) $\frac{1}{4}$

72. $\int \frac{x^{e-1} + e^{x-1}}{x^e + e^x} dx =$

(1) $\frac{1}{e} \log |x^e + e^x| + c$

(2) $e \log |x^e + e^x| + c$

(3) $-\frac{1}{e} \log |x^e + e^x| + c$

(4) $-e \log |x^e + e^x| + c$

73. $\int \frac{x + \sin x}{1 + \cos x} dx =$

(1) $x \tan \frac{x}{2} + \sec \frac{x}{2} + c$

(2) $x \sec \frac{x}{2} + \tan \frac{x}{2} + c$

(3) $x \sin \frac{x}{2} + \cos \frac{x}{2} + c$

(4) $x \tan \frac{x}{2} + c$

74. $\int \frac{5x^2 + 3}{x^2(x^2 - 2)} dx =$

(1) $\frac{13}{4\sqrt{2}} \log \left| \frac{x + \sqrt{2}}{x - \sqrt{2}} \right| + \frac{3}{2x} + c$

(2) $\frac{13}{2\sqrt{2}} \log \left| \frac{\sqrt{2} - x}{\sqrt{2} + x} \right| + \frac{3}{2x} + c$

(3) $\frac{5}{3\sqrt{2}} \log \left| \frac{x + \sqrt{2}}{x - \sqrt{2}} \right| + \frac{3}{5}x + c$

(4) $\frac{13}{4\sqrt{2}} \log \left| \frac{x - \sqrt{2}}{x + \sqrt{2}} \right| + \frac{3}{2x} + c$

75. If $I_n = \int \frac{\sin nx}{\cos x} dx$, then $I_n =$

(1) $\frac{-2}{n-1} \cos(n-1)x - I_{n-2}$

(2) $\frac{-2}{n+1} \sin(n-1)x - I_{n-2}$

(3) $\frac{-2}{n+1} \cos(n-1)x - I_{n-2}$

$$(4) \frac{2}{n-1} \cos(n-1)x - I_{n-2}$$

$$76. \lim_{n \rightarrow \infty} \left[\frac{1^k + 2^k + 3^k + \dots + n^k}{n^{k+1}} \right] =$$

$$(1) \frac{1}{k}$$

$$(2) \frac{2}{k}$$

$$(3) \frac{1}{k+1}$$

$$(4) \frac{2}{k+1}$$

$$77. \text{ If } I = \int_0^{10} f(x) dx = 5, \text{ then } \sum_{k=-1}^{10} \int_0^1 f(k-1+x) dx =$$

$$(1) 50$$

$$(2) 20$$

$$(3) 10$$

$$(4) 5$$

78. The area (in sq. units) enclosed between the curves $y = x^2$ and $y = |x|$ is

(1) $\frac{2}{3}$

(2) 1

(3) $\frac{1}{6}$

(4) $\frac{1}{3}$

79. The differential equations corresponding to the family of circles in the plane touching the Y-axis at the origin is

(1) $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$

(2) $\frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$

(3) $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

(4) $\frac{dy}{dx} = \frac{2xy}{x^2 + y^2}$

80. The solution of the differential equation

$$(x - 4y^3) \frac{dy}{dx} - y = 0 (y > 0)$$

(1) $x + 2y^3 = cy$

(2) $y = x^3 + cx$

(3) $x = y^3 + cy$

(4) $y + 2x^3 = cx$

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81. In the determination of the internal resistance of a cell with a potentiometer, the error in the measurement of the balancing length is ± 1 mm. When the cell alone is connected in the circuit, the balancing length is obtained at 60 cm and when the cell is shunted with a resistance of $10\ \Omega \pm 2\%$, the balancing length is obtained at 50 cm. The error in the determination of the internal resistance of the cell is _____
- (1) 1.8%
 - (2) 2.4%
 - (3) 4.2%
 - (4) 5.6%
82. A bird is tossing (flying to and fro) between two cars moving towards each other on a straight road. One car has speed of 54 kmh^{-1} while the other has the speed of 36 kmh^{-1} . The bird starts moving from first car towards the other and is moving with the speed of 36 kmh^{-1} where the two cars were separated

by 36 km. The total distance covered by the bird before the cars meet each other is

- (1) 1440 m
- (2) 24400 m
- (3) 244 m
- (4) 14400 m

83. A body is projected from the top of a tower with a velocity $\vec{u} = 3\hat{i} + 4\hat{j} + 5\hat{k} \text{ ms}^{-1}$ where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along east, north and vertically upwards respectively. If the height of the tower is 30 m, horizontal range of the body on the ground is

($g = 10 \text{ ms}^{-2}$)

- (1) 12 m
- (2) 9 m
- (3) 25 m
- (4) 15 m

84. Equation of a projectile is given by $y = Px - Qx^2$ where P , Q are constants. The ratio of maximum height to range of the projectile is

(1) $\frac{P^2}{Q}$

(2) $4P$

(3) $\frac{P}{4}$

(4) $\frac{Q^2}{2P}$

85. A person of 60 kg mass is in a lift which is coming down such that the man exerts a force of 150 N on the floor of the lift.

Then the acceleration of the lift is ($g = 10 \text{ ms}^{-2}$)

(1) 40.0 ms^{-2}

(2) 15.0 ms^{-2}

(3) 22.5 ms^{-2}

(4) 7.5 ms^{-2}

86. The situations are shown in fig (a) & (b), in each case, $m_1 = 3 \text{ kg}$ and $m_2 = 4 \text{ kg}$. If a_1, a_2 are the respective

accelerations of the blocks in these situation, then the values of

a_1 and a_2 are respectively $[g = 10 \text{ ms}^{-2}]$

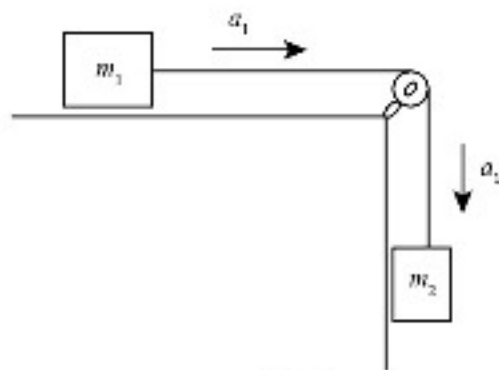


Fig. (a)

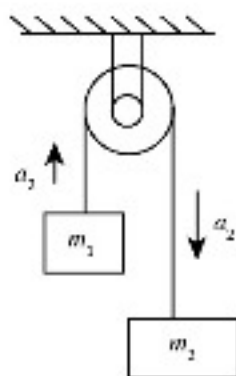


Fig. (b)

(1) $\frac{10}{7} \text{ ms}^{-2}, \frac{25}{7} \text{ ms}^{-2}$

(2) $\frac{30}{7} \text{ ms}^{-2}, \frac{5}{7} \text{ ms}^{-2}$

(3) $\frac{40}{7} \text{ ms}^{-2}, \frac{10}{7} \text{ ms}^{-2}$

(4) $\frac{20}{7} \text{ ms}^{-2}, \frac{10}{7} \text{ ms}^{-2}$

87. A girl of mass 50 kg swinging on a cradle. If she moves with a velocity of 2 ms^{-1} upward in a direction making an angle 60° with the vertical, then the power generated is ($g = 9.8 \text{ ms}^{-2}$)

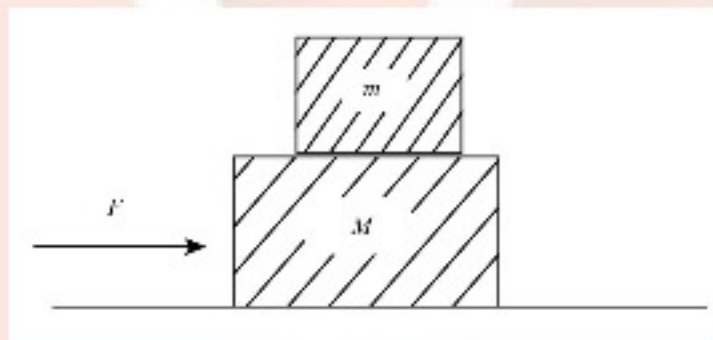
(1) 980 W

(2) $490\sqrt{2}$ W

(3) $490\sqrt{3}$ W

(4) 245 W

88. Two blocks of masses ' M ' and ' m ' are placed on one another on a smooth horizontal surface as shown in the figure. The force ' F ' is acting on the mass ' M ' horizontally during time interval ' t '. Assuming no relative sliding between the blocks, the work done by friction on the blocks is _____



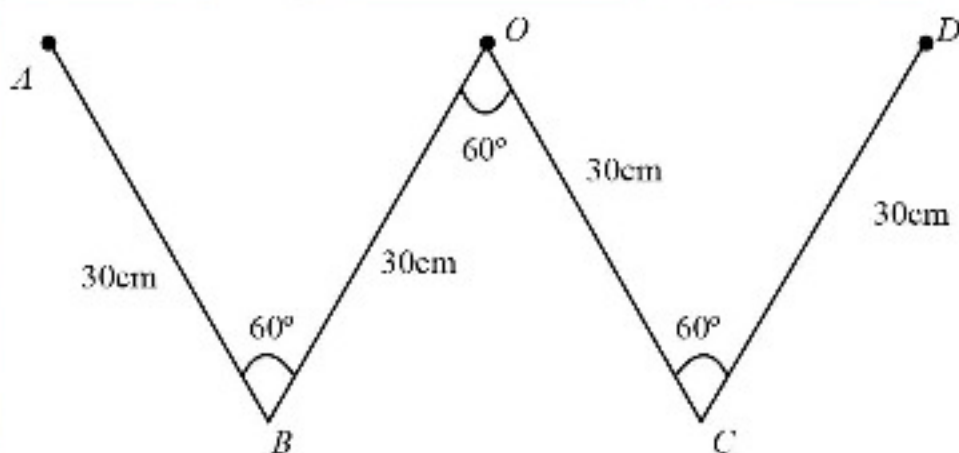
(1) $\frac{mF^2t^2}{2(M+m)^2}$

(2) $\frac{F^2t^2}{(M+m)}$

(3) $\frac{M+m}{mt^2}$

$$(4) \frac{Ft}{2(M+m)}$$

89. A uniform thin rod of 120 cm length 1600 g mass is bent as shown in the figure. The moment of inertia of the bent rod about an axis passing through the point 'O' and perpendicular to the plane of the paper is _____ kg m^2 .



- (1) 0.120
- (2) 0.084
- (3) 0.360
- (4) 0.018

90. A solid sphere is projected up along an inclined plane of inclination 30° with the horizontal with a speed of 4 ms^{-1} . If it rolls without slipping, the maximum distance traversed by it is _____ ($g = 10 \text{ ms}^{-2}$)

- (1) 1.12 m
- (2) 22.4 m
- (3) 112 m
- (4) 2.24 m

91. Time period of a simple pendulum of length ' L ' is T_1 . Time period of a uniform rod of same length ' L ' suspended from one end and oscillating in a vertical plane is T_2 . Amplitude of oscillation is small in both the cases. Then $\frac{T_1}{T_2}$ is

- (1) $\sqrt{\frac{4}{3}}$
- (2) 1
- (3) $\sqrt{\frac{3}{2}}$

(4) $\sqrt{\frac{2}{3}}$

92. Two bodies of masses 4 m and 9 m are separated by a distance 'r'. The gravitational potential at a point on this line joining them where the gravitational field becomes zero is

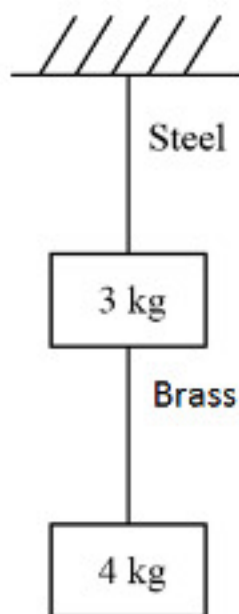
(1) $\frac{-4Gm}{r}$

(2) $\frac{-25Gm}{r}$

(3) $\frac{-9Gm}{r}$

(4) $\frac{-13Gm}{r}$

93. The ratios of lengths, areas of cross-section and Young's moduli of steel to that of brass wires shown in the figure are a , b and c respectively. The ratio of increase in the lengths of brass to that of steel wires is [Assume that the masses of steel and Brass wires are negligible]



- (1) $\frac{4a}{7bc}$
- (2) $\frac{7bc}{4a}$
- (3) $\frac{4ab}{7a}$
- (4) $\frac{7a}{4bc}$

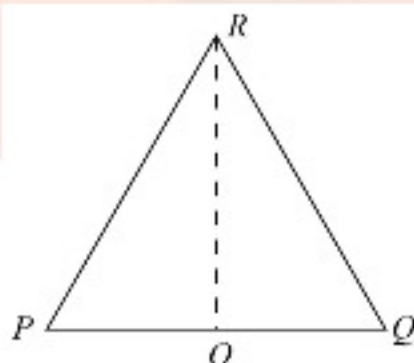
94. Fully filled open water tank has two holes on either side of its walls. One is a square hole of side x cm at a depth of 2 m from the top, and the other hole is equilateral triangle of side 4 cm at

a depth of 6 m from the top. If the rate of flow of water is same from both the holes, then 'x' is

- (1) 12 cm
- (2) 3.46 cm
- (3) 6.92 cm
- (4) 1.73 cm

95. Three uniform thin aluminum rods each of length 2 m form an equilateral triangle PQR as shown in the figure. The mid point of the rod PQ is at the origin of the coordinate system. If the temperature of the system of rods increases by 50°C , the increase in y -coordinate of the centre of mass of the system of the rods is _____ mm.

(Coefficient of volume expansion of aluminium
 $= 12\sqrt{3} \times 10^{-6} \text{ K}^{-1}$)



- (1) 0.05
- (2) 0.1
- (3) 0.2
- (4) 0.8

96. A wall is made of equally thick layers ' P ' and ' Q ' of different materials. Thermal conductivity of ' Q ' is half of that of the ' P '. In the steady state, if the temperature difference across the wall is 24°C , then the temperature difference across the layer ' P ' is

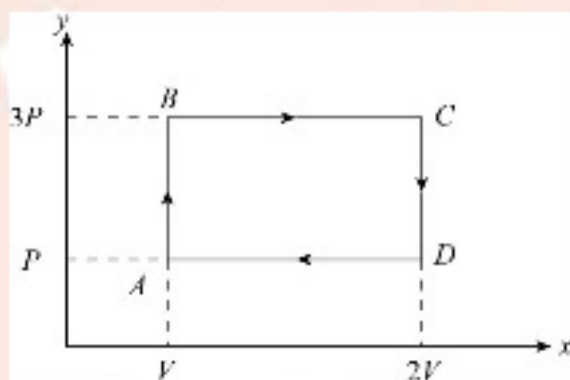
- _____
- (1) 12°C
 - (2) 8°C
 - (3) 16°C
 - (4) 4°C

97. One mole of a gas expands such that its volume ' V ' changes with absolute temperature ' T ' in accordance with the relation $V = KT^2$ where ' K ' is a constant. If the temperature of the gas changes by 60°C , then work done by the gas is (R is universal gas constant).

- (1) $120R$

- (2) $R \ln 60$
- (3) $K R \ln 60$
- (4) $40KR$

98. A monatomic ideal gas goes through a cyclic process as shown in the figure. The efficiency of this process is



- (1) 78%
 - (2) 62%
 - (3) 42%
 - (4) 21%
99. If the average translational kinetic energy of a molecule in a gas is equal to the kinetic energy of an electron accelerating from rest through 10 volt, then the temperature of the gas molecule is (Boltzmann constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$)

- (1) 73.7 K
- (2) 77.3×10^3 K
- (3) 7.73×10^3 K
- (4) 730 K

100. The transverse displacement of a string linear density 0.01 kgm^{-1} , clamped at its ends is given by

$$Y_{(x,t)} = 0.03 \sin\left(\frac{2\pi x}{3}\right) \cos(60\pi t) \text{ where } x \text{ and } y \text{ are in meters}$$

and time ' t ' is in seconds. Tension in the string is

- (1) 81 N
- (2) 162 N
- (3) 36 N
- (4) 9 N

101. A closed organ pipe of length ' L ' and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases is equal in both the pipes. If the frequencies of their first overtones are same, then the length of the open organ pipe is

- (1) $\frac{L}{3}$
- (2) $\frac{4L}{3}$
- (3) $\frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$
- (4) $\frac{4L}{3} \sqrt{\frac{\rho_2}{\rho_1}}$

102. Two point sources S_1 and S_2 are 24 cm apart. Where should a convex lens of focal length 9 cm be placed in between them so that the images of both sources are formed at the same place?

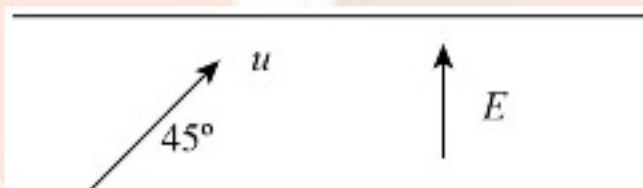
- (1) 8 cm
- (2) 6 cm
- (3) 10 cm
- (4) 12 cm

103. Two slits separated by 0.5 mm are illuminated by light of wavelength 500 nm. The screen is at a distance of 120 cm from the slits. The phase difference between the interfering waves at

a point 3 mm on the screen from the central bright fringe is

- _____
- (1) π
 - (2) 3π
 - (3) 5π
 - (4) 7π

104. The uniform electric field intensity between the two plates of a parallel plate capacitor is $1 \times 10^3 \text{ Vm}^{-3}$ acting vertically upwards as shown in the figure. The plates are sufficiently long and have separation 2 cm. A particle of negative charge $1 \mu\text{C}$ and mass 2 g is projected at an angle 45° with the electric field from the lower plate with a velocity ' u '. The maximum velocity acquired by the particle, if it is not hit the upper plate is



- (1) 0.1 ms^{-1}
- (2) 0.2 ms^{-1}
- (3) 2 ms^{-1}

(4) 1 ms^{-1}

105. An infinitely long thin straight wire has uniform linear charge density of $\frac{1}{3} \text{ Cm}^{-1}$. Then the magnitude of the force acting on a charge $3 \text{ } \mu\text{C}$ situated at a point of 18 cm away from the wire is

$$\left(\frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right)$$

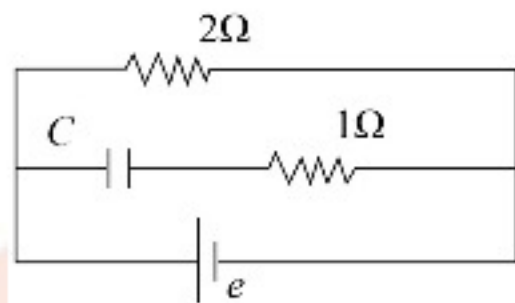
(1) 10^5 N

(2) $2 \times 10^5 \text{ N}$

(3) $\frac{1}{3} \times 10^6 \text{ N}$

(4) $3 \times 10^{11} \text{ N}$

106. In steady state, a capacitor of capacitance $2 \text{ } \mu\text{F}$ is charged to $4 \text{ } \mu\text{C}$, as shown in figure. If internal resistance of the cell is $0.5 \text{ } \Omega$, the emf of the Cell is



- (1) 4 V
- (2) 2.5 V
- (3) 5 V
- (4) 2 V

107. An electrostatic paint sprayer has a metal sphere of diameter 18 cm and at a potential of 25 kV. If it sprays paint droplets, the charge on the metal sphere is

- (1) $0.5 \mu\text{C}$
- (2) $0.25 \mu\text{C}$
- (3) $2.5 \mu\text{C}$
- (4) $25 \mu\text{C}$

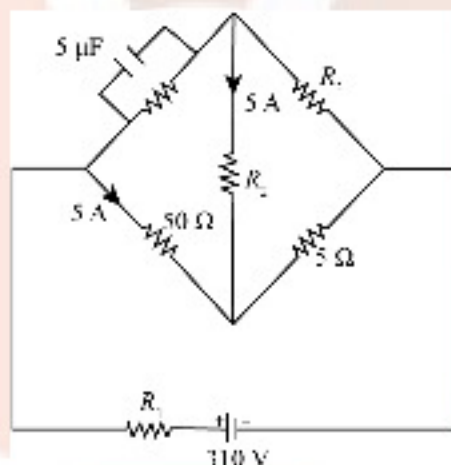
108. When the terminals of a cell are connected by a wire of resistance 4Ω , the potential difference across the cell is 1.6 V.

If a wire of the same resistance is connected in parallel with the first, the potential difference becomes 1.33 V. The emf and internal resistance of the cell are respectively

- (1) 2 V, 2 Ω
- (2) 1 V, 2 Ω
- (3) 1 V, 1 Ω
- (4) 2 V, 1 Ω

109. If the charge on the capacitor is 1 mC in the given circuit, then

$$\frac{R_1 R_2}{R_3} = \text{_____ } \Omega$$



- (1) 0.4
- (2) 10
- (3) 6

(4) 0.6

110. Two concentric coils of 20 turns each are placed in same plane. Their radii are 30 cm and 60 cm and carry 0.4 A and 0.6 A current respectively in opposite directions. The magnetic induction at the centre in Tesla is _____

(1) $\frac{5}{3}\mu_0$

(2) $\frac{8}{3}\mu_0$

(3) $\frac{2}{3}\mu_0$

(4) $\frac{10}{3}\mu_0$

111. Two long parallel conducting wires carrying currents are separated by a distance 'x'. Work done per unit length in changing the distance between the wires is proportional to

(1) x

(2) $\frac{1}{x}$

(3) $\log_e x$

$$(4) \frac{1}{\log_e x}$$

112. Match the following

List - I

- (a) High Retentivity
- (b) High resistivity
- (c) Low coercivity
- (d) Negative susceptibility

List - II

- (i) Telephone diaphragm
- (ii) Diamagnet
- (iii) To decrease eddy currents losses
- (iv) Permanent magnet

The correct answer is

- (1) a-i, b-ii, c-iii, d-iv
- (2) a-iv, b-iii, c-i, d-ii
- (3) a-i, b-iv, c-iii, d-ii
- (4) a-iv, b-ii, c-i, d-iii

113. Assertion (A) : It is more difficult to push magnet into a coil with more number of turns.

Reason (R) : The emf induced in a coil opposes the motion of a magnet when it is moved towards the coil.

- (1) Both A and R are true. R is a correct explanation of A.

- (2) Both A and R are true. R is not correct explanation of A
- (3) A is true, R is false
- (4) A is false, R is true

114. When a coil is connected to AC supply of frequency 50 Hz, a current of 4 A flows in it and it consumes 240 W power. If the potential difference across the coil is 100 V, then the inductance value of the coil is

- (1) $L = \frac{1}{5\pi}$ H
- (2) $L = \frac{\pi}{5}$ H
- (3) $L = (5\pi)$ H
- (4) $L = \frac{1}{25\pi}$ H

115. The magnitude of electric field in an electromagnetic wave is 60 Vm^{-1} . Then the amplitude of magnetic field is

- (1) $2 \times 10^7 \text{ T}$
- (2) $6 \times 10^7 \text{ T}$

(3) $6 \times 10^{-7} \text{ T}$

(4) $2 \times 10^{-7} \text{ T}$

116. Both an electron and a photon have same de-Broglie wavelength of $1.2 \text{ \AA}$. The ratio of their energies is nearly

(1) 1 : 1

(2) 1 : 100

(3) 1 : 100

(4) 1 : 1000

117. The radius of orbit of an electron and the speed of electron in the ground state of hydrogen atom are $5.5 \times 10^{-11} \text{ m}$ and $4 \times 10^6 \text{ m/s}$ respectively. Then, the orbital period of this electron in the first excited state will be _____

(1) $9.608 \times 10^{-16} \text{ s}$

(2) $8.9068 \times 10^{-16} \text{ s}$

(3) $7.806 \times 10^{-16} \text{ s}$

(4) $6.908 \times 10^{-16} \text{ s}$

118. The half life of a stream of radioactive particles moving along a straight path with a constant kinetic energy of 4 eV is 1 minute. The percentage of particles which decay before travelling a distance of 3.6 km is _____

(Mass of the radioactive particles = 3.2×10^{-21} kg and charge of the electron = 1.6×10^{-19} C).

- (1) 37.5
- (2) 87.5
- (3) 75
- (4) 175

119. A Zener diode voltage regulator operated in the range 120–180 V produces a constant supply of 110 V and 250 mA to the load. If the maximum current is equally shared between the load and the Zener diode, then the values of load resistance (R_L) and series resistance (R_S) are respectively

- (1) $R_L = 70 \Omega$, $R_S = 280 \Omega$
- (2) $R_L = 440 \Omega$, $R_S = 140 \Omega$
- (3) $R_L = 440 \Omega$, $R_S = 1400 \Omega$
- (4) $R_L = 280 \Omega$, $R_S = 70 \Omega$

120. The process of recovering the modulating signal from the modulated carrier wave is called

- (1) Amplification
- (2) Rectification
- (3) Noise
- (4) Detection

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121. The energy and radius of electron present in second orbit of He^+ respectively are

- (1) $-2.18 \times 10^{-18} \text{ J}$, 105.8 pm
- (2) $-4.36 \times 10^{-18} \text{ J}$, 52.9 pm
- (3) $-1.09 \times 10^{-18} \text{ J}$, 105.8 pm
- (4) $-8.72 \times 10^{-18} \text{ J}$, 211.6 pm

122. The work functions of Ag, Mg, K and Na respectively in eV are 4.3, 3.7, 2.25 and 2.30. When an electromagnetic radiation of wavelength of 300 nm is allowed to fall on these metal surfaces, the number of metals from which the electrons are ejected is
($1 \text{ eV} = 1.6022 \times 10^{-19} \text{ J}$)

- (1) 5
- (2) 4
- (3) 3
- (4) 2

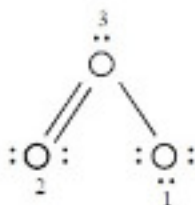
123. Assertion (A) : Na^+ and Mg^{2+} ions are isoelectronic but the ionic radius of Na^+ is greater than that of Mg^{2+}

Reason (R) : The effective nuclear charge of Na^+ ion is less than that of Mg^{2+} ion

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

124. The Lewis structure for O_3 molecule is given below.

The correct formal charges on oxygen atoms labelled 1, 2, 3 are respectively



- (1) +1, 0, -1
- (2) -1, 0, +1

(3) 0, +1, -1

(4) -1, -1, 0

125. Which among the following are having, diamagnetic property?

(a) B_2 (b) N_2 (c) O_2 (d) C_2

(1) a, b

(2) b, c

(3) a, d

(4) b, d

126. At 27°C in a 10 L flask 4.0 g of an ideal gaseous mixture containing He (molar mass 4.0 g mol^{-1}) and Ne (molar mass 20 g mol^{-1}) has a pressure of 1.23 atm. What is the mass % of neon? ($R = 0.082\text{ L atm K}^{-1}\text{ mol}^{-1}$)

(1) 84.2

(2) 25.2

(3) 74.2

(4) 62.5

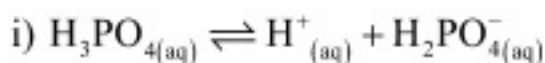
127. Two oxides of a non-metal X contains 50% and 40% of non-metal respectively. If the formula of the first oxide is XO_2 , then the formula of second oxide is

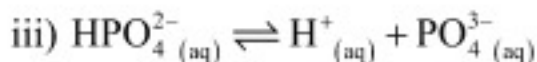
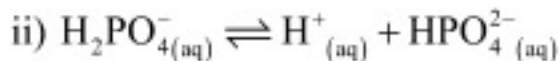
- (1) XO_3
- (2) X_2O
- (3) X_2O_3
- (4) X_2O_5

128. The enthalpy of formation ($\Delta_f H$) of methanol, formaldehyde and water -239 , -116 and -286 kJ mol^{-1} respectively. The enthalpy change for the oxidation of methanol to formaldehyde and water in kJ is

- (1) -136
- (2) -173
- (3) -163
- (4) 163

129.





The equilibrium constants for the above reactions at a certain temperature are K_1 , K_2 and K_3 respectively. The equilibrium constant for the reaction $\text{H}_3\text{PO}_{4(\text{aq})} \rightleftharpoons 3\text{H}^+{}_{(\text{aq})} + \text{PO}_4^{3-}{}_{(\text{aq})}$ in terms of K_1 , K_2 and K_3 is

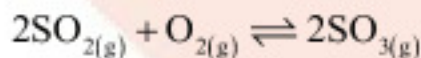
(1) $K_1 + K_2 + K_3$

(2) $\frac{K_1}{K_2 + K_3}$

(3) $\frac{K_3}{K_1 K_2}$

(4) $K_1 K_2 K_3$

130. The equilibrium constant (K_c) for the following equilibrium



at 563 K is 100. At equilibrium, the number of moles of SO_3 in the 10 litre flask is twice the number of moles of SO_2 , then the number of moles of oxygen is

(1) 0.3

(2) 0.4

(3) 0.1

(4) 0.2

131. How many milliliters of 20 volume H_2O_2 solution is needed to react completely with 500 mL of acidified 1N KMnO_4 solution?

(1) 56

(2) 140

(3) 224

(4) 280

132. Which one of the following statements is **not** correct?

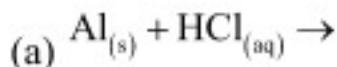
(1) Alkali metals react with water liberating oxygen gas

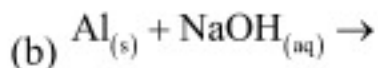
(2) The hydration enthalpies of alkali metals ions decrease down the group

(3) KO_2 is paramagnetic

(4) Lithium halides are somewhat Covalent in nature

133. In which of the following reactions, hydrogen is liberated?





(1) a, c

(2) a, b

(3) b, c

(4) a, b, c

134. Which one of the following statements is **not** correct?

(1) In CO molecule, there are one 'sigma' (σ) and two "Pi" (π) bonds.

(2) Both $[\text{SiF}_6]^{2-}$ and $[\text{SiCl}_6]^{2-}$ ions are known

(3) In CO_2 molecule, carbon hybridisation is sp.

(4) Fullerenes are made by heating graphite in an electric arc in the presence of argon gas.

135. Which of the following forms holes in the ozone layer?

(1) SO_2

(2) CO_2

(3) CO

(4) CF_2Cl_2

136. Identify the correct statements from the following

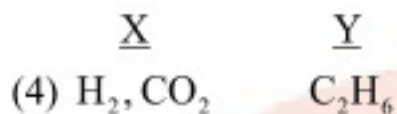
- a) Electromeric effect is a permanent effect
- b) Hyper conjugation is a temporary effect
- c) Fractional distillation is used to separate two liquids from a mixture if the difference in their boiling point is less
- d) Different compounds are adsorbed on an adsorbent to different extents

The correct answer is

- (1) c, d
- (2) a, b, c
- (3) b, d
- (4) b, c, d

137. The gaseous products formed at cathode (X) and anode (Y), when an aqueous solution of sodium acetate is electrolysed are

- | | |
|--|------------------------------------|
| <u>X</u> | <u>Y</u> |
| (1) $\text{C}_2\text{H}_6, \text{H}_2$ | CO_2 |
| <u>X</u> | <u>Y</u> |
| (2) CO_2 | $\text{C}_2\text{H}_6, \text{H}_2$ |



138. Identify the correct statements for a ring system to exhibit aromaticity?

- a) It must not be planar
- b) It must possess $(4n + 2)\pi$ electrons.
- c) It must be planar
- d) It must possess $4n \pi$ electrons

The correct answer is

- (1) a, b
- (2) b, c
- (3) a, d
- (4) b, d

139. An element has a body centered cubic structure with a unit cell edge length of 400 pm. Atomic mass of an element is 24 g mol^{-1} . What is the density of the element?

$$(N_A = 6 \times 10^{23} \text{ mol}^{-1})$$

(1) 3.60 g cm^{-3}

(2) 1.80 g cm^{-3}

(3) 1.25 g cm^{-3}

(4) 2.50 g cm^{-3}

140. 31 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) is dissolved in 600 g of water. The freezing point depression of the solution is (K_f for water is $1.86 \text{ K kg mol}^{-1}$)

(1) 3.10 K

(2) 4.65 K

(3) 0.77 K

(4) 1.55 K

141. Which one of the following solutions of compounds show highest osmotic pressure? (AB , AB_2 and A_2B_3 are ionic compounds)

(1) 3.0 M AB type $i = 1.6$ and temperature is 27°C

(2) 2.5 M AB_2 type $i = 2.5$ and temperature is 57°C

(3) 1.5 M A_2B_3 type $i = 4.1$ and temperature is 27°C

(4) 5.0 M urea $i = 1.0$ and temperature is 67°C

142. 20% of a first order reaction was found to be completed at 10 A.M. At 11.30 A.M. on the same day, 20% of the reaction was found to be remaining. The half life period in minutes of the reaction is

- (1) 90
- (2) 60
- (3) 45
- (4) 30

143. Same amount of electricity is passed through aqueous solutions of AgNO_3 and CuSO_4 . The number of Ag and Cu atoms deposited are x and y respectively. The correct relationship between x and y is

- (1) $x = y$
- (2) $x = 2y$
- (3) $y = 2x$
- (4) $x < y$

144. Colloidal solution of gold is in different colours like red, purple, blue and golden because of

- (1) Size difference in the particles of gold
- (2) Variable oxidation states of gold
- (3) Presence of impurities
- (4) Difference in the concentration of gold particles

145. Copper matte contains

- (1) CuO , Cu_2S
- (2) CuO , FeS
- (3) Cu_2S , FeS
- (4) Cu_2S , FeO

146. $\text{S} + \text{Conc. H}_2\text{SO}_4 \rightarrow \text{X} + \text{Y}$

Here X is a gas and Y is a liquid and both are triatomic molecules. The number of electron lone pairs present on the central atoms of X and Y are respectively

- (1) 2, 1
- (2) 2, 2

(3) 1, 2

(4) 1, 0

147. Which of the following is used in the estimation of carbon monoxide?

(1) Cl_2O_7

(2) BrO_3

(3) I_2O_4

(4) I_2O_5

148. To prepare XeF_6 , Xe and F_2 are mixed at 573 K and 60-70 bar in the ratio of

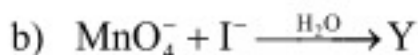
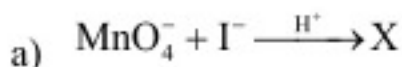
(1) 1:20

(2) 20:1

(3) 1:5

(4) 5:1

149. What are X and Y in the following reactions ?



- (1) IO_3^- , I_2
- (2) I_2 , IO_3^-
- (3) IO_3^- , IO_4^-
- (4) I_2 , IO_4^-

150. Identify from the following, the diamagnetic, tetrahedral complex

- (1) $[\text{Ni}(\text{CN})_4]^{2-}$
- (2) $[\text{Ni}(\text{Cl})_4]^{2-}$
- (3) $[\text{Ni}(\text{CO})_4]$
- (4) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

151. Which one of the following is not used as an initiator in ionic polymerisation?

- (1) AlCl_3
- (2) SnCl_2
- (3) NaNH_2



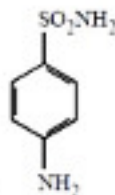
152. Identify the statements which are **not** correct from the following

- a) Carbohydrates are stored as glycogen in animals
- b) In glycylalanine, $-\overset{\text{O}}{\parallel}{\text{C}}-$ of peptide bond belongs to alanine
- c) Base – Sugar – phosphate unit is known as nucleoside
- d) Obesity is due to hypothyroidism

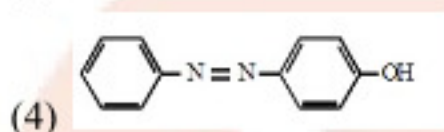
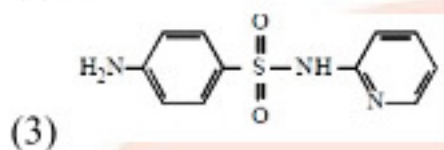
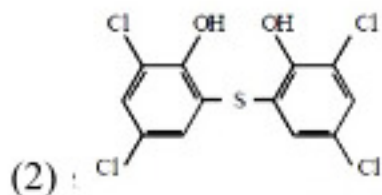
The correct answer is

- (1) a, c, d
- (2) a, d
- (3) b, c
- (4) b, c, d

153. Which of the following structure represents the compound, generally added to soaps to impart antiseptic properties?



(1)



154. The drug, which was designed to prevent the interaction of histamine with the receptors present in the stomach wall is:

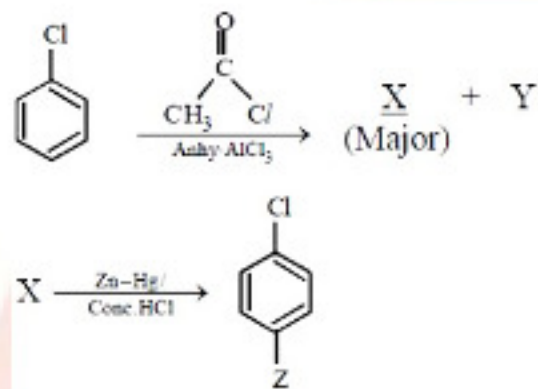
- (1) Equanil
- (2) Prontosil
- (3) Aspartame
- (4) Cimetidine

155. Which one of the following is more reactive towards S_N2 reaction?

- (1) $(CH_3)_3CX$
- (2) $(CH_3)_2CHX$
- (3) CH_3CH_2X

(4) CH_3X

156. What is Z in the following sequence of reactions?



(1) $-\text{CH}_3$

(2) $-\text{CH}_2\text{CH}_3$

(3) $\begin{array}{c} -\text{C}-\text{Cl} \\ || \\ \text{O} \end{array}$

(4) $\begin{array}{c} -\text{C}-\text{CH}_3 \\ || \\ \text{O} \end{array}$

157. Identify the statements which is **not** correct?

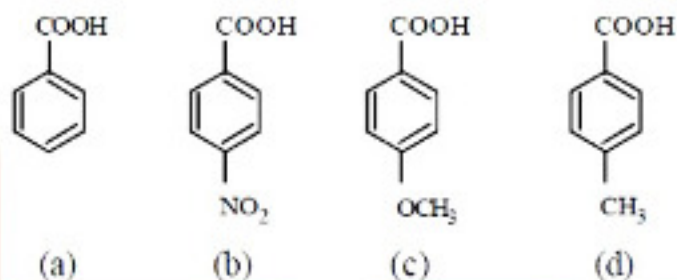
(1) Freon 12 is manufactured by Swartz reaction

(2) CHCl_3 is stored in closed, dark coloured bottles

(3) Chronic exposure to CHCl_3 causes liver damage

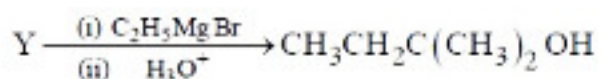
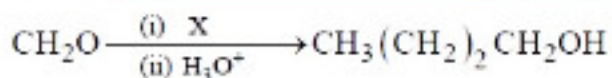
- (4) Dehydrobromination of 2-bromopentane gives pent-1-ene as the major product

158. The increasing order of acidity of the following carboxylic acids is



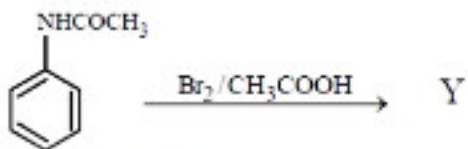
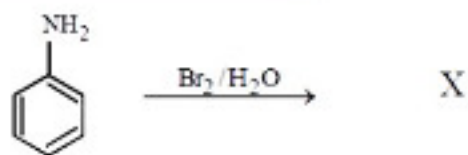
- (1) $a < b < d < c$
- (2) $c < d < a < b$
- (3) $c < d < b < a$
- (4) $b < c < d < a$

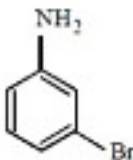
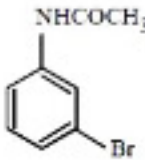
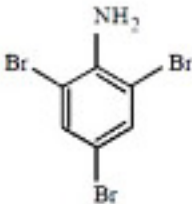
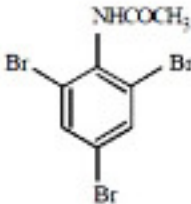
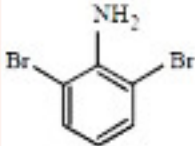
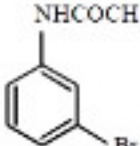
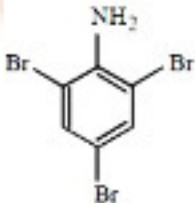
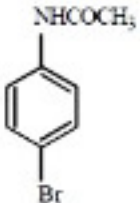
159. What are X and Y in the following reactions?



- | | <u>X</u> | <u>Y</u> |
|-----|---|---|
| (1) | $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{MgBr}$ | $\text{C}_2\text{H}_5\text{COCH}_3$ |
| (2) | $\text{CH}_3 - \text{CH}_2 - \text{MgBr}$ | $\text{CH}_3\text{CH}_2\text{CHO}$ |
| (3) | $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$ | $\text{CH}_3 - \text{CO} - \text{CH}_3$ |
| (4) | $(\text{CH}_3)_3\text{CMgBr}$ | $\text{CH}_3 - \text{CO} - \text{CH}_3$ |

160. What are X and Y in the following reactions?



- (1) X Y
-  
- (2) X Y
-  
- (3) X Y
-  
- (4) X Y
-  

AP EAMCET-2017-Engineering-26 Apr-Shift-1
Answer Key

1	3	33	4	65	4	97	1	129	4
2	3	34	4	66	2	98	4*	130	2
3	4	35	1	67	1	99	2	131	3
4	2	36	3	68	3	100	1	132	1
5	4	37	4	69	1	101	3	133	4
6	2	38	4	70	1	102	2	134	2
7	1	39	4	71	4	103	3	135	4
8	3	40	4	72	1	104	2	136	4
9	2	41	4	73	4	105	1	137	3
10	2	42	3	74	1	106	2	138	2
11	2	43	2	75	1	107	2	139	3
12	1	44	3	76	3	108	4	140	4
13	3	45	4	77	4	109	4	141	2
14	2	46	4	78	4	110	4	142	3
15	3	47	4	79	3	111	3	143	2
16	1	48	2	80	1	112	2	144	1
17	4	49	3	81	3	113	1	145	3

18	3	50	3	82	4	114	1	146	3
19	1	51	1	83	4	115	4	147	4
20	4	52	4	84	3	116	3	148	1
21	3	53	3	85	4	117	4	149	2
22	1	54	4	86	3	118	2	150	3
23	3	55	3	87	3	119	2	151	4
24	1	56	1	88	1	120	4	152	3
25	4	57	3	89	2	121	1	153	2
26	1	58	2	90	4	122	3	154	4
27	2	59	1	91	3	123	1	155	4
28	1	60	4	92	2	124	2	156	2
29	4	61	1	93	3	125	4	157	4
30	1	62	2	94	2	126	4	158	2
31	1	63	2	95	3	127	1	159	3
32	1	64	4	96	2	128	3	160	4

AP EAMCET-2017-Engineering-26 Apr-Shift-1
Solutions – Mathematics

1. Consider the function.

$$f(x) = \begin{cases} 3x - 1 & \text{for } x > 3 \\ x^2 + 1 & \text{for } -3 \leq x \leq 3 \\ 2x - 3 & \text{for } x < -3 \end{cases}$$

So,

$$\begin{aligned} f(-4) &= 2(-4) - 3 \\ &= -11 \end{aligned}$$

$$\begin{aligned} f(-2) &= (-2)^2 + 1 \\ &= 4 + 1 \\ &= 5 \end{aligned}$$

$$\begin{aligned} f(-1) &= (-1)^2 + 1 \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} f(0) &= 0^2 + 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} f(3) &= 3^2 + 1 \\ &= 9 + 1 \\ &= 10 \end{aligned}$$

And,

$$\begin{aligned} f(5) &= 3(5) - 1 \\ &= 15 - 1 \\ &= 14 \end{aligned}$$

Therefore, the range of the function is $\{-11, 5, 2, 1, 10, 14\}$

2. Consider the function.

$$\sigma(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ -\frac{(n-1)}{2}, & \text{if } n \text{ is odd} \end{cases}$$

When n is even.

$$\sigma(n) = 1, 2, 3, 4, 5, \dots$$

When n is odd

$$\sigma(n) = 0, -1, -2, -3, -4, -5, \dots$$

The range of $\sigma(n) = Z$

Thus, range of $\sigma(n) = \text{codomain of } \sigma(n) = Z$

Thus, $\sigma(n)$ is one-one and onto function.

3. Let $S_n = \frac{1}{2 \cdot 5} + \frac{1}{5 \cdot 8} + \frac{1}{8 \cdot 11} + \dots n \text{ terms}$

It is simplified as,

$$\begin{aligned} S_n &= \frac{1}{3} \left[\frac{3}{2 \cdot 5} + \frac{8-5}{5 \cdot 8} + \frac{11-8}{8 \cdot 11} + \dots n \text{ terms} \right] \\ &= \frac{1}{3} \left[\frac{5-2}{2 \cdot 5} + \frac{3}{5 \cdot 8} + \frac{3}{8 \cdot 11} + \dots n \text{ terms} \right] \\ &= \frac{1}{3} \left[\left(\frac{1}{2} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{8} \right) + \left(\frac{1}{8} - \frac{1}{11} \right) + \dots \left(\frac{1}{3n-1} - \frac{1}{3n+2} \right) \right] \\ &= \frac{1}{3} \left[\frac{1}{2} - \frac{1}{3n+2} \right] \end{aligned}$$

Solve further,

$$\begin{aligned} S_n &= \frac{1}{3} \left[\frac{3n+2-2}{2(3n+2)} \right] \\ &= \frac{n}{2(3n+2)} \end{aligned}$$

4. The given system of equation can be written as,

$$(a-1)x - y - z = 0$$

$$x - (b-1)y + z = 0$$

$$x + y - (c-1)z = 0$$

The system of above equation is non-trivial if,

$$\begin{vmatrix} a-1 & -1 & -1 \\ 1 & -(b-1) & 1 \\ 1 & 1 & -(c-1) \end{vmatrix} = 0$$

$$\begin{vmatrix} a-1 & -1 & -1 \\ 1 & 1-b & 1 \\ 1 & 1 & 1-c \end{vmatrix} = 0$$

$$(a-1)(1-b)(1-c) - a + 1 - c - b = 0$$

$$ab + bc + ca = abc$$

5. The given matrix form of system of three linear equations in three unknown is,

$$AX = D$$

The system of linear equations will be inconsistent if rank of A is not equal to the rank of AD and rank of AD is greater than rank of A .

Since, the rank of augmented matrix is greater than or equal to the rank of coefficient matrix.

Thus,

$$\frac{\text{rank of } A}{\text{rank of } AD} < 1$$

6. It is given that,

$$\Delta = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ -1 & -\cos \theta & 1 \end{vmatrix}$$

Apply operation $R_3 \rightarrow R_3 + R_1$.

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ 0 & 0 & 2 \end{vmatrix} \\ &= 2(1 + \cos^2 \theta) \end{aligned}$$

Now,

$$0 \leq \cos^2 \theta \leq 1$$

$$1 \leq 1 + \cos^2 \theta \leq 2$$

$$2 \leq 2(1 + \cos^2 \theta) \leq 4$$

Thus,

$$\Delta \in [2, 4]$$

7. Consider the function.

$$\left| z + \frac{2}{z} \right| \geq \left| |z| - \left| \frac{2}{z} \right| \right|$$

It is simplified as,

$$\begin{aligned}\left|z + \frac{2}{z}\right| &\geq \left|z - \frac{2}{z}\right| \\ &\geq \left|z + 2\left(\frac{-1}{z}\right)\right|\end{aligned}$$

Also,

$$\begin{aligned}|z| &\geq 5 \\ -|z| &\leq -5 \\ \frac{-1}{|z|} &\leq \frac{-1}{5}\end{aligned}$$

Therefore,

$$\begin{aligned}\left|z + \frac{2}{z}\right| &\geq \left|5 - \frac{2}{5}\right| \\ &\geq \frac{23}{5}\end{aligned}$$

8. Consider the expression.

$$\log_{\frac{1}{\sqrt{3}}} \left\{ \frac{|z|^2 - |z| + 1}{2 + |z|} \right\} > -2$$

$$\frac{|z|^2 - |z| + 1}{2 + |z|} < \left(\frac{1}{\sqrt{3}} \right)^{-2}$$

$$|z|^2 - |z| + 1 < (\sqrt{3})^2 (2 + |z|)$$

$$|z|^2 - |z| + 1 < 6 + 3|z|$$

Solve further,

$$|z|^2 - 4|z| + 1 < 6$$

$$(|z| - 2)^2 < 9$$

$$-3 < |z| - 2 < 3$$

$$-1 < |z| < 5$$

Therefore,

$$0 < |z| < 5$$

Hence, z lies inside the circle.

9. Consider the expression.

$$\begin{aligned}K &= \sum_{r=1}^{n-1} r(r+1-w)(r+1-w^2) \\&= \sum_{r=1}^{n-1} r((r+1)^2 - (r+1)(w+w^2) + w^3) \\&= \sum_{r=1}^{n-1} r((r+1)^2 - (r+1)(-1)+1) \\&= \sum_{r=1}^{n-1} (r^3 + 3r^2 + 3r)\end{aligned}$$

Solve further,

$$\begin{aligned}K &= \sum_{r=1}^{n-1} r^3 + 3 \sum_{r=1}^{n-1} r^2 + 3 \sum_{r=1}^{n-1} r \\&= \left(\frac{(n-1)n}{2} \right)^2 + \frac{3(n-1)n(2n-1)}{6} + \frac{3(n-1)n}{2} \\&= \frac{n^2(n-1)^2}{4} + \frac{(n-1)n(2n-1)}{2} + \frac{3(n-1)n}{2} \\&= \frac{n(n-1)}{4} [n^2 + 3n + 4]\end{aligned}$$

10. It is given that α is a non real root of $x^7 = 1$

Thus,

$$\alpha^7 = 1, \alpha \neq 1$$

Now,

$$\begin{aligned}\alpha(1+\alpha)(1+\alpha^2+\alpha^4) &= \alpha(1+\alpha^2+\alpha^4+\alpha+\alpha^3+\alpha^5) \\ &= \alpha+\alpha^2+\alpha^5+\alpha^2+\alpha^4+\alpha^6 \\ &= \frac{\alpha(1-\alpha^6)}{1-\alpha} \\ &= \frac{\alpha-\alpha^7}{1-\alpha}\end{aligned}$$

Solve further,

$$\begin{aligned}\alpha(1+\alpha)(1+\alpha^2+\alpha^4) &= \frac{\alpha-1}{1-\alpha} \\ &= -1\end{aligned}$$

11. Consider the given inequation,

$$\frac{x-1}{3x+4} - \frac{x-3}{3x-2} < 0$$

It is simplified as,

$$\frac{(x-1)(3x-2)-(x-3)(3x+4)}{(3x+4)(3x-2)} < 0$$

$$\frac{(3x^2 - 2x - 3x + 2) - (3x^2 + 4x - 9x - 12)}{(3x+4)(3x-2)} < 0$$

$$\frac{-5x + 2 + 5x + 12}{(3x+4)(3x-2)} < 0$$

$$\frac{14}{(3x+4)(3x-2)} < 0$$

Solve further,

$$(3x+4)(3x-2) < 0$$

$$x \in \left(-\frac{4}{3}, \frac{2}{3}\right)$$

12 Let the numbers are a and b .

$$\frac{2ab}{a+b} = -\frac{8}{5}$$

And,

$$\sqrt{ab} = 2$$

So,

$$\frac{2 \times 4}{a+b} = -\frac{8}{5}$$

$$a+b = -5$$

So,

$$\begin{aligned}(2a)(2b) &= 4ab \\ &= 4 \times 4 \\ &= 16\end{aligned}$$

And,

$$\begin{aligned}2a + 2b &= 2(a + b) \\ &= 2(-5) \\ &= -10\end{aligned}$$

Hence, the required quadratic equation is,

$$x^2 + 10x + 16 = 0$$

13. Let the roots of the equation be $\frac{a}{r}, a, ar$,

So,

$$\frac{a}{r} + a + ar = 7$$

And,

$$\frac{a}{r} \cdot a + a \cdot ar + ar \cdot \frac{a}{r} = 14$$

And,

$$\frac{a}{r} \cdot a \cdot ar = 8$$

Thus,

$$a = 2$$

Therefore,

$$\frac{2}{r} + 2r = 5$$

$$2r^2 - 5r + 2 = 0$$

$$2r(r - 2) - 1(r - 2) = 0$$

$$(r - 2)(2r - 1) = 0$$

$$\text{Thus, } r = 2, \frac{1}{2}$$

Hence, the roots are 1, 2, 4

Therefore, the required difference is,

$$\text{Diff} = 1, 2, 4$$

14. The roots of the equation are α, β, γ .

So,

$$\alpha + \beta + \gamma = -p$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

$$\alpha\beta\gamma = -r$$

Now,

$$\begin{aligned}
 (1+\alpha^2)(1+\beta^2)(1+\gamma^2) &= (1+\alpha^2)(1+\beta^2+\gamma^2+(\beta\gamma)^2) \\
 &= \left[1+\beta^2+\gamma^2+(\beta\gamma)^2+\alpha^2+(\alpha\beta)^2 \right] \\
 &\quad \left[+(\alpha\gamma)^2+(\alpha\beta\gamma)^2 \right] \\
 &= \left[1+(\alpha^2+\beta^2+\gamma^2)+ \right. \\
 &\quad \left. ((\beta\gamma)^2+(\alpha\beta)^2+(\alpha\gamma)^2)+(\alpha\beta\gamma)^2 \right] \\
 &= \left[1+\left[(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha) \right]+ \right. \\
 &\quad \left. \left[(\alpha\beta+\beta\gamma+\gamma\alpha)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma) \right] \right]
 \end{aligned}$$

Simplify further,

$$\begin{aligned}
 (1+\alpha^2)(1+\beta^2)(1+\gamma^2) &= 1 + [p^2 - 2q] + [q^2 - 2rp] + r^2 \\
 &= 1 + p^2 - 2q + q^2 - 2rp + r^2 \\
 &= (q-1)^2 + (r-p)^2
 \end{aligned}$$

15. The total number of ways of making a garland such that no two red roses come together is calculated as,

$$\begin{aligned}
 \text{Ways} &= \frac{6 \times 5!}{2} = \frac{720 \times 120}{2} \\
 &= 43200
 \end{aligned}$$

16. It is given that there are 10 intermediate stations between A and B . There are 7 stations where the trains does not stop and the three stations where the train stops.

Therefore, the total number of ways is,

$$\begin{aligned}\text{Number of ways} &= {}^8 C_3 \\ &= \frac{8 \times 7 \times 6}{1 \times 2 \times 3} \\ &= 56\end{aligned}$$

17. Consider the given expansion.

$$\begin{aligned}x &= {}^{37} C_4 + \sum_{r=1}^5 {}^{42-r} C_r \\ &= {}^{37} C_4 + {}^{41} C_1 + {}^{40} C_2 + {}^{39} C_3 + {}^{38} C_3 + {}^{37} C_3 \\ &= {}^{37} C_4 + {}^{37} C_4 + {}^{38} C_3 + {}^{39} C_3 + {}^{40} C_3 + {}^{41} C_3 \\ &= {}^{38} C_4 + {}^{38} C_3 + {}^{39} C_3 + {}^{40} C_3 + {}^{41} C_3\end{aligned}$$

Solve further,

$$\begin{aligned}x &= {}^{40} C_4 + {}^{40} C_3 + {}^{41} C_3 \\ &= {}^{41} C_4 + {}^{41} C_3 \\ &= {}^{42} C_4\end{aligned}$$

18. The coefficient of x^r in the expansion of $(1+x)^n$ is given by nC_r .

Therefore, the coefficient of x^5 in the given expansion is calculated as,

$$\begin{aligned}\text{Coeff } x^5 &= {}^{21}C_5 + {}^{22}C_5 + \dots + {}^{30}C_5 \\ &= ({}^{21}C_6 + {}^{21}C_5 + {}^{22}C_5 + \dots + {}^{30}C_5) - {}^{21}C_6 \\ &= ({}^{22}C_6 + {}^{22}C_5 + \dots + {}^{30}C_5) - {}^{21}C_6\end{aligned}$$

Expanding in the similar pattern further,

$$\begin{aligned}\text{Coeff } x^5 &= ({}^{31}C_6 + {}^{30}C_5) - {}^{21}C_6 \\ &= {}^{31}C_6 - {}^{21}C_6\end{aligned}$$

19. Consider the expression.

$$x = \frac{1 \cdot 3}{3 \cdot 6} + \frac{1 \cdot 3 \cdot 5}{3 \cdot 6 \cdot 9} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{3 \cdot 6 \cdot 9 \cdot 12} + \dots$$

The above expression is solved as,

$$\begin{aligned}
 x &= \frac{1 \cdot 3}{3^2 \cdot (2!)} + \frac{1 \cdot 3 \cdot 5}{3^3 (3!)} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{3^4 (4!)} + \dots \\
 &= \frac{\frac{1}{2} \left(\frac{1}{2} + 1 \right)}{2!} \left(\frac{2}{3} \right)^2 + \frac{\frac{1}{2} \left(\frac{1}{2} + 1 \right) \left(\frac{1}{2} + 2 \right)}{3!} \left(\frac{2}{3} \right)^3 + \dots \\
 &= \left[1 + \frac{1}{2} \left(\frac{2}{3} \right) + \frac{\frac{1}{2} \left(\frac{1}{2} + 1 \right)}{2!} \left(\frac{2}{3} \right)^2 \right. \\
 &\quad \left. + \frac{\frac{1}{2} \left(\frac{1}{2} + 1 \right) \left(\frac{1}{2} + 2 \right)}{3!} \left(\frac{2}{3} \right)^3 + \dots \right] + \left(1 + \frac{1}{3} \right) \\
 &= \left(1 - \frac{2}{3} \right)^{-1/2} - \frac{4}{3}
 \end{aligned}$$

Solve further,

$$\begin{aligned}
 x &= \left(\frac{1}{3} \right)^{-1/2} - \frac{4}{3} \\
 &= \sqrt{3} - \frac{4}{3}
 \end{aligned}$$

$$3x + 4 = 3\sqrt{3}$$

$$(3x + 4)^2 = 3(\sqrt{3})^2$$

Solve further,

$$9x^2 + 24x + 16 = 27$$

$$9x^2 + 24x = 11$$

20. Consider the given equation.

$$\frac{5x^2 + 2}{x^3 + x} = \frac{A_1}{x} + \frac{A_2x + A_3}{x^2 + 1}$$

It is solved as,

$$\frac{5x^2 + 2}{x(x^2 + 1)} = \frac{A_1(x^2 + x) + (A_2x + A_3)x}{x(x^2 + 1)}$$

$$5x^2 + 2 = A_1(x^2 + 1) + A_2x^2 + A_3x$$

Compare the coefficient of like terms.

$$A_1 + A_2 = 5$$

$$A_3 = 0$$

$$A_1 = 2$$

Therefore,

$$A_1 = 2$$

$$A_2 = 3$$

$$A_3 = 0$$

21. Consider the quadratic equations.

$$x^2 + px + q = 0$$

Its roots are,

$\tan \alpha$ and $\tan \beta$

So,

$$\tan \alpha + \tan \beta = -p$$

And,

$$\tan \alpha \cdot \tan \beta = q$$

So,

$$\begin{aligned}\tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{-p}{1 - q} \\ &= \frac{p}{q - 1}\end{aligned}$$

Now,

$$\begin{aligned}L &= \sin^2(\alpha + \beta) + p \cos(\alpha + \beta) \sin(\alpha + \beta) + q \cos^2(\alpha + \beta) \\ &= \cos^2(\alpha + \beta) [\tan^2(\alpha + \beta) + p \tan(\alpha + \beta) + q] \\ &= \frac{1}{\sec^2(\alpha + \beta)} [\tan^2(\alpha + \beta) + p \tan(\alpha + \beta) + q] \\ &= \frac{1}{1 + \tan^2(\alpha + \beta)} [\tan^2(\alpha + \beta) + p \tan(\alpha + \beta) + q]\end{aligned}$$

Solve further,

$$\begin{aligned} L &= \frac{1}{1 + \frac{p^2}{(q-1)^2}} \left[\frac{p^2}{(q-1)^2} + \frac{p^2}{q-1} + q \right] \\ &= \frac{(q-1)^2}{(q-1)^2 + p^2} \left[\frac{p^2 + p^2(q-1)^2 + q(q-1)^2}{(q-1)^2} \right] \\ &= \frac{q \{ p^2 + (q-1)^2 \}}{\{ p^2 + (q-1)^2 \}} \\ &= q \end{aligned}$$

22. Consider the expression.

$$(1 + \cos 10^\circ) + (\cos 20^\circ + \cos 30^\circ)$$

It is solved as,

$$\begin{aligned} (1 + \cos 10^\circ) + (\cos 20^\circ + \cos 30^\circ) &= 2 \cos^2 5^\circ + 2 \cos 25^\circ \cos 5^\circ \\ &= 2 \cos 5^\circ (\cos 5^\circ + \cos 25^\circ) \\ &= 2 \cos 5^\circ (2 \cos 15^\circ \cos 10^\circ) \\ &= 4 \cos 5^\circ \cos 15^\circ \cos 10^\circ \end{aligned}$$

23. Consider the trigonometric equations.

$$1 + \cos x \cdot \cos 5x = \sin^2 x$$

It is simplified as,

$$1 - \sin^2 x + \cos x \cdot \cos 5x = 0$$

$$\cos^2 x + \cos x \cdot \cos 5x = 0$$

$$\cos x (\cos x + \cos 5x) = 0$$

$$\cos x (2 \cos 3x \cos 2x) = 0$$

Therefore,

$$\cos x = 0$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

Also,

$$\cos 3x = 0$$

$$3x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \frac{11\pi}{2}$$

And,

$$\cos 2x = 0$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

Thus, the solution are,

$$x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Hence, there are 10 solutions.

24. It is given that,

$$\cot(\cos^{-1} x) = \sec \left\{ \tan^{-1} \left(\frac{a}{\sqrt{b^2 - a^2}} \right) \right\}$$

The above expression is simplified as,

$$\cot \left(\cot^{-1} \frac{x}{\sqrt{1-x^2}} \right) = \sec \left\{ \sec^{-1} \sqrt{1 + \left(\frac{a}{\sqrt{b^2 - a^2}} \right)^2} \right\}$$

$$\frac{x}{\sqrt{1-x^2}} = \sqrt{1 + \left(\frac{a}{\sqrt{b^2 - a^2}} \right)^2}$$

$$\frac{x}{\sqrt{1-x^2}} = \sqrt{\frac{b^2 - a^2 + a^2}{b^2 - a^2}}$$

$$\frac{x}{\sqrt{1-x^2}} = \frac{b}{\sqrt{b^2 - a^2}}$$

Square both sides,

$$\frac{x^2}{1-x^2} = \frac{b^2}{b^2 - a^2}$$

$$x^2 b^2 - x^2 a^2 = b^2 - b^2 x^2$$

$$x^2 (2b^2 - a^2) = b^2$$

$$x = \frac{b}{\sqrt{2b^2 - a^2}}$$

25. Consider the expression.

$$y = \operatorname{sech}^{-1}\left(\frac{1}{\sqrt{2}}\right) + \operatorname{cosech}^{-1}(-1)$$

It is solved as,

$$\begin{aligned} y &= \log_e \left(\frac{1 + \sqrt{1 - 1/2}}{1/\sqrt{2}} \right) + \log_e \left(\frac{1 - \sqrt{1 + 1}}{-1} \right) \\ &= \log_e \frac{\left(\frac{\sqrt{2} + 1}{\sqrt{2}} \right)}{1/\sqrt{2}} + \log_e (\sqrt{2} - 1) \\ &= \log_e \left((\sqrt{2})^2 - 1 \right) \\ &= \log_e (2 - 1) \end{aligned}$$

Solve further,

$$\begin{aligned} y &= \log_e 1 \\ &= 0 \end{aligned}$$

26. Consider $b \cos(C + \theta) + c \cos(B - \theta)$.

It is simplified as,

$$\begin{aligned} b \cos(C + \theta) + c \cos(B - \theta) &= \begin{bmatrix} b(\cos C \cos \theta - \sin C \sin \theta) + \\ c(\cos B \cos \theta + \sin B \sin \theta) \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta (b \cos C + c \cos B) - \\ \sin \theta (b \sin C - c \sin B) \end{bmatrix} \end{aligned}$$

By projection formula and sine rule,

$$\frac{b}{\sin B} = \frac{c}{\sin C}$$

$$b \sin C - c \sin B = 0$$

Therefore,

$$\begin{aligned} b \cos(C + \theta) + c \cos(B - \theta) &= a \cos \theta - \sin \theta \cdot 0 \\ &= a \cos \theta \end{aligned}$$

27. It is given that

$$\cos \theta = \frac{c - a}{b}$$

So

$$\begin{aligned} \sin \theta &= \sqrt{1 - \cos^2 \theta} \\ &= \sqrt{1 - \frac{(c - a)^2}{b^2}} \\ &= \frac{\sqrt{b^2 - (c - a)^2}}{b} \end{aligned}$$

And,

$$\tan \theta = \frac{\sqrt{b^2 - (c-a)^2}}{c-a}$$

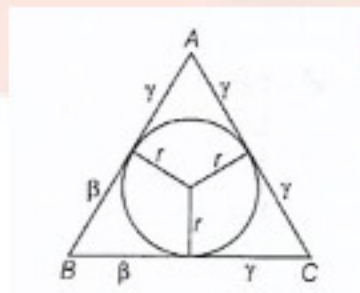
$$\begin{aligned}(c-a) \tan \theta &= \sqrt{b^2 - (c-a)^2} \\ &= \sqrt{-(c^2 + a^2 - b^2) + 2ac} \\ &= \sqrt{2ac - 2ac \cos B}\end{aligned}$$

Solve further,

$$\begin{aligned}(c-a) \tan \theta &= \sqrt{2ac} \sqrt{1 - \cos B} \\ &= \sqrt{2ac} \sqrt{2 \sin^2 \frac{B}{2}} \\ &= 2\sqrt{ac} \sin \frac{B}{2}\end{aligned}$$

28. It is given that α, β, γ are the lengths of tangents from vertices of a triangle to its incircle semi perimeter of $\triangle ABC$.

Consider the figure shown below.



So,

$$S = \alpha + \beta + \gamma$$

And,

$$\begin{aligned}\text{Area of } \Delta ABC &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{(\alpha + \beta + \gamma)(\alpha)(\beta)(\gamma)}\end{aligned}$$

Also,

$$\begin{aligned}r &= \frac{\Delta}{s} \\ &= \frac{\sqrt{(\alpha + \beta + \gamma)(\alpha)(\beta)(\gamma)}}{\alpha + \beta + \gamma}\end{aligned}$$

$$r^2 = \frac{(\alpha + \beta + \gamma)(\alpha)(\beta)(\gamma)}{(\alpha + \beta + \gamma)^2}$$

$$(\alpha + \beta + \gamma)^2 = \frac{(\alpha + \beta + \gamma)(\alpha)(\beta)(\gamma)}{r^2}$$

Solve further,

$$\alpha + \beta + \gamma = \frac{\alpha\beta\gamma}{r^2}$$

29. It is given that,

$$\begin{aligned}(3\bar{i} - \bar{j} + 2\bar{k}) &= \begin{pmatrix} (2\bar{i} + 2\bar{j} - \bar{k})x + (\bar{i} - 2\bar{j} + 2\bar{k})y + \\ (-2\bar{i} + \bar{j} - 2\bar{k})z \end{pmatrix} \\ &= \begin{bmatrix} \bar{i}(2x + y - 2z) + \bar{j}(3x - 2y + z) \\ + \bar{k}(-x + 2y - 2z) \end{bmatrix}\end{aligned}$$

Compare the coefficient of like terms.

$$2x + y - 2z = 3$$

$$3x - 2y + z = -1$$

$$-x + 2y - 2z = 2$$

On solving the three equations,

$$x = 2$$

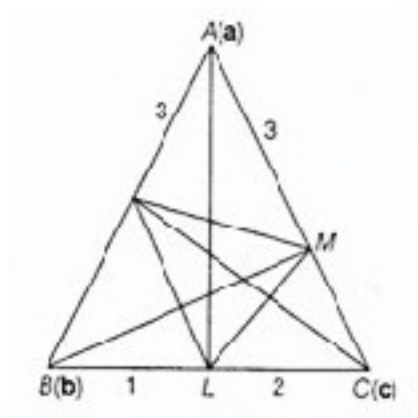
$$y = 5$$

$$z = 3$$

Therefore,

$$(x, y, z) = (2, 5, 3)$$

30. Consider the figure shown below.



The points L, M, N divide the line BC, CA, AB in the ratios $1:2, 2:3$ and $3:5$.

So,

$$L = \frac{2\bar{b} + \bar{c}}{3}$$

$$M = \frac{2\bar{a} + 3\bar{c}}{5}$$

$$N = \frac{3\bar{b} + 5\bar{a}}{8}$$

$$K = \frac{5\bar{b} + 3\bar{a}}{8}$$

So,

$$\begin{aligned}\overline{AL} &= \frac{2\overline{b} + \overline{c}}{3} - \overline{a} \\ &= \frac{2\overline{b} + \overline{c} - 3\overline{a}}{3}\end{aligned}$$

And,

$$\begin{aligned}BM &= \frac{2\overline{a} + 3\overline{c}}{5} - \overline{b} \\ &= \frac{2\overline{a} + 3\overline{c} - 5\overline{b}}{5}\end{aligned}$$

And,

$$\begin{aligned}\overline{CN} &= \frac{3\overline{b} + 5\overline{a}}{8} - \overline{c} \\ &= \frac{3\overline{b} + 5\overline{a} - 8\overline{c}}{8}\end{aligned}$$

And,

$$\begin{aligned}\overline{CK} &= \frac{5\overline{b} + 3\overline{a}}{8} - \overline{c} \\ &= \frac{5\overline{b} + 3\overline{a} - 8\overline{c}}{8}\end{aligned}$$

Therefore,

$$\begin{aligned}
 \left| \frac{\overline{AL} + \overline{BM} + \overline{CN}}{\overline{CK}} \right| &= \left| \frac{\left\{ \left(\frac{2\overline{b} + \overline{c} - 3\overline{a}}{3} \right) + \left(\frac{2\overline{a} + 3\overline{c} - 5\overline{b}}{5} \right) \right\} + \left(\frac{3\overline{b} + 5\overline{a} - 8\overline{c}}{8} \right)}{\left(\frac{5\overline{b} + 3\overline{a} - 8\overline{c}}{8} \right)} \right| \\
 &= \left| \frac{5\overline{b} + 3\overline{a} - 8\overline{c}}{15(5\overline{b} + 3\overline{a} - 8\overline{c})} \right| \\
 &= \frac{1}{15}
 \end{aligned}$$

31. The normal to the plane P_1 is calculated as,

$$\begin{aligned}
 n_1 &= \begin{vmatrix} \overline{i} & \overline{j} & \overline{k} \\ 0 & 1 & -1 \\ 0 & 3 & -2 \end{vmatrix} \\
 &= \overline{i}
 \end{aligned}$$

The normal to the plane P_2 is calculated as,

$$\begin{aligned} n_2 &= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 3 & 0 \\ 1 & -3 & -0 \end{vmatrix} \\ &= -9\bar{k} \end{aligned}$$

So,

$$\begin{aligned} n_1 \times n_2 &= \bar{i} \times (-9\bar{k}) \\ &= -9(\bar{i} \times \bar{k}) \\ &= -9\bar{j} \end{aligned}$$

Therefore, the angle between $-9\bar{j}$ and $\bar{i} + \bar{j} + \bar{k}$ is

$$\begin{aligned} \cos \theta &= \pm \left(\frac{9}{9\sqrt{3}} \right) \\ &= \pm \frac{1}{\sqrt{3}} \\ \theta &= \cos^{-1} \left(\pm \frac{1}{\sqrt{3}} \right) \end{aligned}$$

32. It is given that,

$$\bar{r} \times \bar{b} = \bar{c} \times \bar{b}$$

So,

$$\vec{a} \times (\vec{r} \times \vec{b}) = \vec{a} \times (\vec{c} \times \vec{b})$$

$$(\vec{a} \cdot \vec{b}) \cdot \vec{r} - (\vec{a} \cdot \vec{r}) \cdot \vec{b} = \vec{a} \times (\vec{c} \times \vec{b})$$

$$(\vec{a} \cdot \vec{b}) \cdot \vec{r} = \vec{a} \times (\vec{c} \times \vec{b})$$

$$\vec{r} = \frac{\vec{a} \times (\vec{c} \times \vec{b})}{(\vec{a} \cdot \vec{b})}$$

Solve further,

$$\vec{r} = -\frac{[(\vec{c} \times \vec{b}) \times \vec{a}]}{(\vec{a} \cdot \vec{b})}$$

$$= -\frac{[-(\vec{b} \times \vec{c}) \times \vec{a}]}{(\vec{a} \cdot \vec{b})}$$

$$= \frac{[(\vec{b} \times \vec{c}) \times \vec{a}]}{(\vec{a} \cdot \vec{b})}$$

33. The volume of the tetrahedron formed by the coterminal edges $\vec{a}$, $\vec{b}$ and $\vec{c}$.

$$V = \frac{1}{6} [\bar{a} \bar{b} \bar{c}]$$

$$4 = \frac{1}{6} [\bar{a} \bar{b} \bar{c}]$$

$$[\bar{a} \bar{b} \bar{c}] = 24$$

The volume of parallelepiped formed by the coterminous edges $\bar{a} \times \bar{b}$, $\bar{b} \times \bar{c}$ and $\bar{c} \times \bar{a}$ is,

$$V_{pp} = [\bar{a} \times \bar{b} \bar{b} \times \bar{c} \bar{c} \times \bar{a}]$$

$$= [\bar{a} \bar{b} \bar{c}]^2$$

$$= 24^2$$

$$= 576$$

34. It is given that the points are coplanar.

$$\overline{AB} = -\bar{i} + 5\bar{j} - 3\bar{k}$$

$$\overline{BC} = -3\bar{i} - 2\bar{j} + 6\bar{k}$$

$$\overline{CD} = 5\bar{i} + 4\bar{j} + (\lambda - 2)\bar{k}$$

So,

$$|\overline{AB} \ \overline{BC} \ \overline{CD}| = 0$$

$$\begin{vmatrix} -1 & 5 & -3 \\ -3 & -2 & 6 \\ 5 & 4 & \lambda - 2 \end{vmatrix} = 0$$

$$2\lambda + 20 + 15(\lambda - 2) + 150 + 36 - 30 = 0$$

$$17\lambda + 146 = 0$$

Solve further,

$$\lambda = -\frac{146}{17}$$

35. Construct the table as shown below.

x_i	f_i	$d_i = x_i - a$	d_i^2	$f_i d_i$	$f_i d_i^2$
6	2	-12	144	-24	288
10	4	-8	64	-32	256
14	7	-4	16	-28	112
$18 = a$	12	0	0	0	0
24	8	6	36	48	288
28	4	10	100	40	400
30	3	12	144	36	432
	$N = \sum f_i = 40$			40	1776

The variance is calculated as,

$$\begin{aligned}\text{Variance} &= \frac{1}{N} \sum f_i d_i^2 - \left(\frac{\sum f_i d_i}{\sum f_i} \right)^2 \\ &= \frac{1}{40} (1776) - 1 \\ &= \frac{1736}{40} \\ &= 434\end{aligned}$$

36. The median of the given observation is calculated as,

$$\begin{aligned}\text{Median} &= \frac{25\alpha + 26\alpha}{2} \\ &= 25.5\alpha\end{aligned}$$

The mean deviation about the median is calculated as,

$$\begin{aligned}\text{Mean deviation about median} &= \frac{\sum_{k=1}^{50} |k\alpha - 25.5\alpha|}{50} \\ 50 \times 50 &= \sum_{k=1}^{50} |\alpha(k - 25.5)| \\ 2500 &= |\alpha| \sum_{k=1}^{50} |k - 25.5|\end{aligned}$$

$$z|\alpha|[24.5 + 23.5 + \dots + 15 + 0.5] = 2500$$

Solve further,

$$2|\alpha| \times \frac{25}{2} \times (24.5 + 0.5) = 2500$$

$$|\alpha| \times 25 = 1400$$

$$|\alpha| = 4$$

37. The total number of possible outcomes is $^{10}C_3$

Let A be the event that minimum of chosen number is 3 and B be the event that maximum of chosen number is 7.

Number of ways of choosing remaining two numbers from A is,

$$\begin{aligned} n(A) &= {}^7C_2 \\ &= 21 \end{aligned}$$

Number of ways of choosing remaining two numbers from B is,

$$\begin{aligned} n(B) &= {}^6C_2 \\ &= 15 \end{aligned}$$

Therefore, the required probability is,

$$\begin{aligned} P &= \frac{21 + 15 - 3}{120} \\ &= \frac{33}{120} \\ &= \frac{11}{40} \end{aligned}$$

38. Let E_1 be the event that $1W$ and $3B$ marbles are transferred, E_2 be the event $2W$ and $2B$ marbles are transferred, E_3 be the event that $3W$ and $1B$ marbles are transferred and E_4 be the event that $4W$ and $0B$ marbles are transferred.

Let A be the event that a black marble is drawn from the bag.

Therefore, the required probability is,

$$\begin{aligned}
 P(E_1|A) &= \frac{P(E_1) \cdot P(A|E_1)}{\left(P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + \right. \\
 &\quad \left. P(E_3) \cdot P(A|E_3) + P(E_4) \cdot P(A|E_4) \right)} \\
 &= \frac{\frac{5}{{}^8C_4} \cdot \frac{3}{4}}{\frac{5}{{}^8C_4} \cdot \frac{3}{4} + \frac{30}{{}^8C_4} \cdot \frac{2}{4} + \frac{30}{{}^8C_4} \cdot \frac{1}{4} + 0} \\
 &= \frac{15}{15 + 60 + 30} \\
 &= \frac{1}{7}
 \end{aligned}$$

39. It is clear from the given information.

$$\begin{aligned}\alpha &= P(X < 3) \\ &= P(X = 1) + P(X = 2) \\ &= 3\lambda\end{aligned}$$

And,

$$\begin{aligned}\beta &= P(X > 2) \\ &= P(X = 3) + P(X = 4) \\ &= 7\lambda\end{aligned}$$

Therefore,

$$\begin{aligned}\alpha : \beta &= 3\lambda : 7\lambda \\ &= 3 : 7\end{aligned}$$

40. Let $X \sim B(n, p)$

So,

$$np = \frac{4}{3}$$

And,

$$npq = \frac{8}{9}$$

Therefore,

$$q = \frac{2}{3}$$

$$p = \frac{1}{3}$$

$$n = 4$$

Hence,

$$\begin{aligned}P(X = 2) &= {}^4C_2 p^2 q^2 \\&= 6 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^2 \\&= \frac{24}{81} \\&= \frac{8}{27}\end{aligned}$$

41. It is given that $p, x_1, x_2, \dots, x_n$ and $q, y_1, y_2, \dots, y_n$ are in AP and their common difference are a and b .

$$x_n = p + na$$

$$x_1 = p + a$$

And,

$$y_n = q + nb$$

$$y_1 = q + b$$

Now, α is the AM of $x_1, x_2, \dots, x_n$

$$\begin{aligned}\alpha &= \frac{n}{2} \left(\frac{x_1 + x_n}{n} \right) \\ &= \frac{x_1 + x_n}{2}\end{aligned}$$

And, β is the AM of $y_1, y_2, \dots, y_n$

$$\beta = \frac{y_1 + y_n}{2}$$

Therefore,

$$\begin{aligned}\alpha &= \frac{p + a + p + na}{2} \\ &= \frac{2p + a(n+1)}{2}\end{aligned}$$

And,

$$\begin{aligned}\beta &= \frac{q + a + q + na}{2} \\ &= \frac{2q + b(n+1)}{2}\end{aligned}$$

Hence,

$$\begin{aligned}\frac{2\alpha - 2p}{a} &= \frac{2\beta - 2q}{b} \\ b(\alpha - p) &= a(\beta - q)\end{aligned}$$

Thus, the locus of the point $p(\alpha, \beta)$ is $b(x - p) = a(y - q)$.

42. Compare the given equation with,

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$a = 2$$

$$h = 2$$

$$b = -6$$

$$g = 1$$

And,

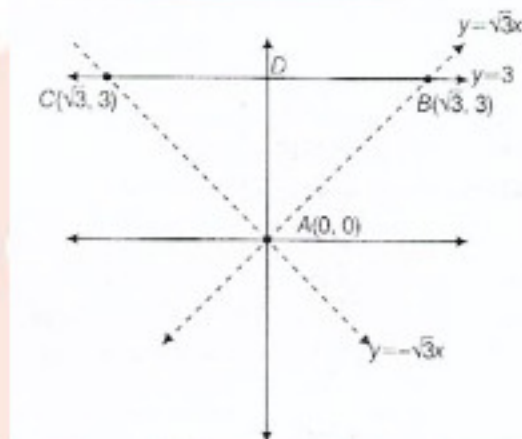
$$f = 4$$

$$c = 1$$

Therefore, the required points are,

$$\begin{aligned}\text{Required Points} &= \left(\frac{bg - fh}{h^2 - ab}, \frac{af - gh}{h^2 - ab} \right) \\ &= \left(\frac{-6 - 8}{4 + 12}, \frac{8 - 2}{4 + 12} \right) \\ &= \left(\frac{-14}{16}, \frac{6}{16} \right) \\ &= \left(\frac{-7}{8}, \frac{3}{8} \right)\end{aligned}$$

43. Consider the triangle shown below.



The triangle is formed by the given lines as shown in the figure.

The triangle ABC is an isosceles triangle.

Therefore, the incentre lies on the median to base.

Also D is the midpoint of BC . Hence, OD is the median to the base BC .

Therefore, the incentre lies on Y-axis.

44. The points $(1, 2)$ and $(3, 4)$ lie on the same side of line,

$$3x - 5y + a = 0$$

Therefore,

$$3(1) - 5(2) + a < 0$$

$$3(3) - 5(4) + a < 0$$

Or,

$$3(1) - 5(2) + a > 0$$

$$3(3) - 5(4) + a > 0$$

So,

$$a < 7, a < 11$$

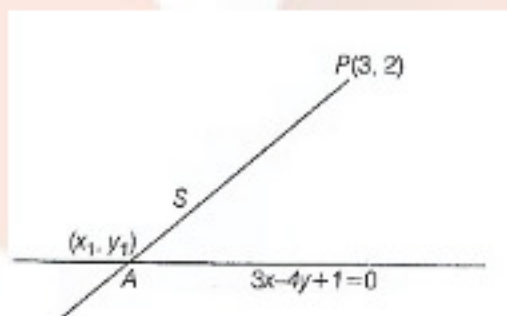
Or,

$$a > 7, a > 11$$

Therefore, the set of all values of a is,

$$\{x \in \mathbb{R} : x > 11\} \cup \{x \in \mathbb{R} : x < 7\}$$

45. Consider the figure shown below.



$$\frac{x_1 - 3}{\cos \theta} = \frac{y_1 - 2}{\sin \theta} = \pm 5$$

So,

$$x_1 = 3 \pm 5 \cos \theta$$

And,

$$y_1 = 2 \pm 5 \sin \theta$$

Now, the points (x_1, y_1) lies on the line $3x - 4y - 1 = 0$.

$$3(3 \pm 5 \cos \theta) - 4(2 \pm 5 \sin \theta) - 1 = 0$$

$$9 \pm 15 \cos \theta - 8 \pm 20 \sin \theta - 1 = 0$$

$$15 \cos \theta - 20 \sin \theta = 0$$

$$3 \cos \theta = \pm 4 \sin \theta$$

Solve further,

$$\tan \theta = \pm \frac{3}{4}$$

So,

$$\cos \theta = \pm \frac{4}{5}$$

$$\sin \theta = \pm \frac{3}{5}$$

Hence,

$$\begin{aligned} x_1 &= 3 \pm \left(\frac{4}{5} 5 \right) \\ &= 7, -1 \end{aligned}$$

And,

$$y_1 = 2 \pm 5\left(\frac{3}{5}\right) \\ = 5, -1$$

Therefore, the coordinates are $(7, 5)$ and $(-1, -1)$.

46. Consider the equation.

$$4xy + 2x + 6y + 3 = 0$$

It can be written as,

$$2x(2y + 1) + 3(2y + 1) = 0$$

$$(2y + 1)(2x + 3) = 0$$

The equation of lines are,

$$x = -\frac{3}{2}$$

And,

$$y = -\frac{1}{2}$$

The equation of line passing through $(2, 1)$ and perpendicular

to the line $x = \frac{23}{2}$ and $y = \frac{1}{2}$ are,

$$y - 1 = 0$$

$$x - 2 = 0$$

Thus,

$$(x-2)(y-1) = 0$$

$$xy - x - 2y + 2 = 0$$

47. Consider the given equations,

$$x^2 - 16pxy - y^2 = 0$$

And,

$$x^2 - 16xy - y^2 = 0$$

So, the equations of bisectors of these lines are,

$$-8px^2 - 2xy + 8py^2 = 0$$

$$4px^2 + xy - 4py^2 = 0$$

And,

$$-8qx^2 - 2xy + 8qy^2 = 0$$

$$4qx^2 + xy - 4qy^2 = 0$$

Therefore,

$$\frac{1}{4q} = \frac{-16p}{1} = \frac{-1}{-4q}$$

So,

$$1 = -64pq$$

$$pq = \frac{-1}{64}$$

48. The equation of pair of lines $6x^2 + 13xy + 6y^2 = 0$ are,

$$3x + 2y = 0$$

And,

$$2x + 3y = 0$$

The equation of pair of lines

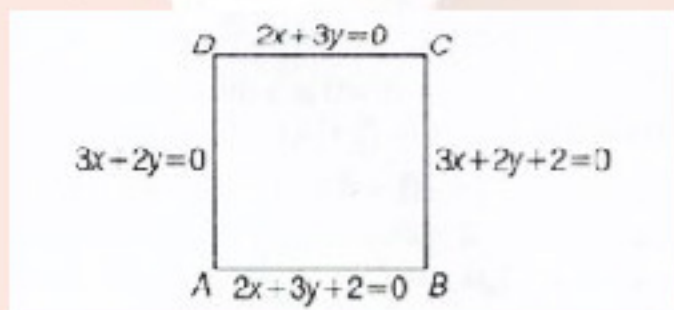
$$6x^2 + 13xy + 6y^2 + 10x + 10y + 4 = 0 \text{ are,}$$

$$3x + 2y + 2 = 0$$

And,

$$2x + 3y + 2 = 0$$

Consider the figure shown below.



AB is parallel to CD and BC is parallel to AD .

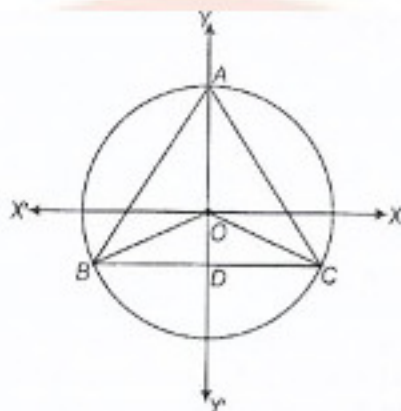
So, $ABCD$ is a parallelogram.

Also,

$$AB = BC = CD = AD = \frac{2}{\sqrt{13}}$$

Therefore, $ABCD$ is a rhombus.

49. Consider the figure shown below.



Therefore,

$$\begin{aligned}AO &= \frac{2}{3} AD \\&= \frac{2}{3} \times 9 \\&= 6\end{aligned}$$

Now, O is the circumcentre of $\triangle ABC$.

In an equilateral triangle circumcentre, incentrem centroid coincide.

Therefore,

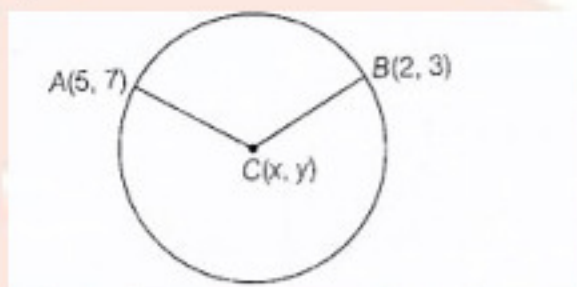
AO is the radius of the circle.

Thus, the equation of circle is,

$$x^2 + y^2 = 6^2$$

$$x^2 + y^2 = 36$$

50. Consider the figure shown below.



Here, $C(x, y)$ is the centre of the circle.

Thus,

$$CA = CB$$

$$CA^2 = CB^2$$

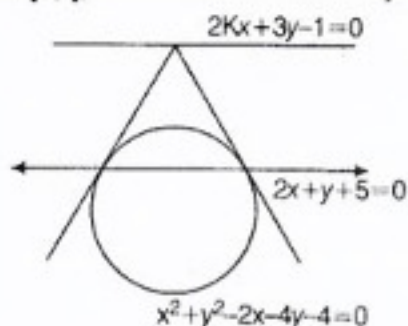
$$(5 - x)^2 + (7 - y)^2 = (2 - x)^2 + (3 - y)^2$$

$$74 - 13 = 6x + 8y$$

Simplify further,

$$6x + 8y = 61$$

51. Consider the figure shown below.



Let (x_1, y_1) lie on the line $2kx + 3y - 1 = 0$

The equation of the chord to the circle is,

$$x^2 + y^2 - 2x - 4y - 4 = 0$$

And,

$$xx_1 + yy_1 - \frac{2(x + x_1)}{2} - \frac{4(y + y_1)}{2} - 4 = 0$$

$$x(x_1 - 1) + y(y_1 - 2) - x_1 - 2y_1 - 4 = 0$$

The lines $2x + y + 5 = 0$ and $2kx + 3y + 1 = 0$ are conjugate lines.

So,

$$2x + y + 5 = 0$$

And

$$x(x_1 - 1) + y(y_1 - 2) - x_1 - 2y_1 - 4 = 0 \text{ coincides.}$$

Therefore,

$$\frac{2}{x_1 - 1} = \frac{1}{y_1 - 2} = \frac{-5}{x_1 + 2y_1 + 4} = \frac{1}{\lambda}$$

So,

$$x_1 = 2\lambda + 1$$

$$y_1 = \lambda + 2$$

$$x_1 + y_1 + 4 = -5\lambda$$

On solving the above three equations.

$$\lambda = -1$$

$$x_1 = -1$$

$$y_1 = 1$$

Thus, the value of k is obtained as,

$$2kx + 3y - 1 = 0$$

$$-2k + 3 - 1 = 0$$

$$k = 1$$

52. The chord of contact of two tangents drawn from the point $P(a, b)$ to the given circle is

$$xa + yb - (x + a) - 3(y + b) - 8 = 0$$

$$(a - 1)x + (b - 3)y - (a + 3b + 8) = 0$$

This line coincides with $5x + y + 1 = 0$

Therefore,

$$\frac{a-1}{5} = \frac{b-3}{1} = \frac{-(a+3b+8)}{1}$$

So,

$$a-1 = 5(a+3b+8)$$

$$6a+15b = -39$$

$$2a+5b = -13$$

And,

$$b-3 = -a-3b-8$$

$$a+4b = -5$$

On solving the equations,

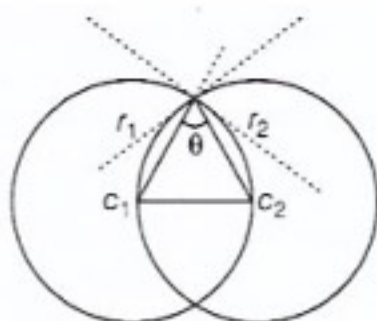
$$a = -9$$

$$b = 1$$

Therefore,

$$\begin{aligned} 5a+b &= 5(-9)+1 \\ &= -45+1 \\ &= -44 \end{aligned}$$

53. Consider the figure shown below.



$$\begin{aligned}\cos \theta &= \frac{r_1^2 + r_2^2 - c_1 c_2}{2r_1 r_2} \\ &= \frac{r_1^2 + r_2^2 - r_1^2}{2r_1 r_2} \\ &= \frac{1}{2} \\ \theta &= \frac{\pi}{3}\end{aligned}$$

Therefore, the angle between the two circle is

$$\theta = \frac{\pi}{3}, \frac{2\pi}{3}$$

54. Consider the given circles,

$$S_1 \equiv x^2 + y^2 + 2x + 2y + 1 = 0$$

And,

$$S_2 \equiv x^2 + y^2 + 4x + 6y + 4 = 0$$

The equation of common chord of these two circles,

$$s_1 - s_2 = 0$$

$$-2x - 4y - 3 = 0$$

$$2x + 4y + 3 = 0$$

The equation of the required circle is,

$$S_1 + \lambda(s_1 - s_2) = 0$$

$$(x^2 + y^2 + 2x + 2y + 1) + \lambda(2x + 4y + 3) = 0$$

$$x^2 + y^2 + 2x(1 + \lambda) + 2y(2\lambda + 1) + 3\lambda + 1 = 0$$

Now, $2x + 4y + 3 = 0$ is the diameter of the circle.

So,

$$2(-(1 + \lambda)) + 4(-(2\lambda + 1)) + 3 = 0$$

$$-2 - 2\lambda - 8\lambda - 4 + 3 = 0$$

$$10\lambda = -3$$

$$\lambda = \frac{-3}{10}$$

Therefore, the equation of the required circle is,

$$x^2 + y^2 + 2x\left(1 - \frac{3}{10}\right) + 2y\left(\frac{-6}{10} + 1\right) - \frac{9}{10} + 1 = 0$$

$$x^2 + y^2 + \frac{14x}{10} + \frac{8y}{10} + \frac{1}{10} = 0$$

$$10x^2 + 10y^2 + 14x + 8y + 1 = 0$$

55. The general equation of parabola is given by,

$$x = Ay^2 + By + C \quad (A \neq 0)$$

The parabola passes through $(-2, 0)$, $(1, 2)$ and $(-1, 3)$.

$$3A + B = 3$$

$$5A + B = -2$$

On solving the above equation.

$$A = \frac{-5}{2}$$

$$B = \frac{21}{2}$$

And,

$$C = -10$$

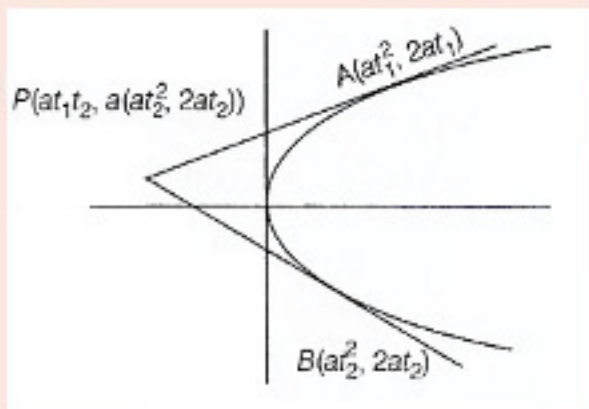
Therefore, the required equation of parabola is,

$$x = \frac{-5}{2}y^2 + \frac{21}{2}y - 10$$

$$2x = -5y^2 + 21y - 20$$

$$5y^2 + 2x - 21y + 20 = 0$$

56. Consider the figure shown below.



P is intersecting the tangent at $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$.

The coordinate of point P is $P(at_1t_2, a(t_1 + t_2))$

The slope of the line PA is,

$$\begin{aligned}\tan \theta_1 &= \frac{2at_1 - at_1 - at_2}{at_1^2 - at_1t_2} \\ &= \frac{1}{t_1}\end{aligned}$$

The slope of line PB is,

$$\begin{aligned}\tan \theta_2 &= \frac{2at_2 - at_1 - at_2}{at_2^2 - at_1t_2} \\ &= \frac{1}{t_2}\end{aligned}$$

Now,

$$\tan \theta_1 + \tan \theta_2 = b$$

$$\frac{1}{t_1} + \frac{1}{t_2} = b$$

$$t_2 + t_1 = bt_1t_2$$

$$\frac{y}{a} = \frac{bx}{a}$$

Solve further,

$$y = bx$$

Therefore, P lies on the line $y = bx$.

57. The equation of tangents to the ellipse can be rewritten as,

$$y = 2xw + \sqrt{76}$$

Its slope is

$$m_1 = 2$$

And,

$$y = -\frac{1}{2}x + 4$$

Its slope is,

$$m_2 = -\frac{1}{2}$$

So,

$$\begin{aligned} m_1 \times m_2 &= 2 \times \left(\frac{-1}{2} \right) \\ &= -1 \end{aligned}$$

The tangents are perpendicular to each other.

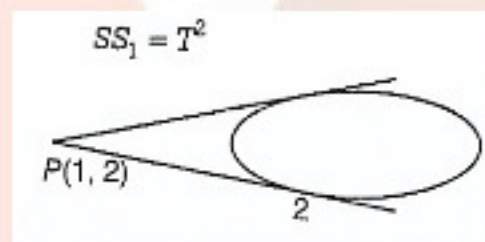
Hence, the required circle is,

$$x^2 + y^2 = a^2 + b^2$$

$$x^2 + y^2 = 16 + 12$$

$$x^2 + y^2 = 28$$

58. Consider the figure shown below.



Let $S = 3x^2 + 2y^2 - 5$.

The equation of pair of tangent drawn to the ellipse from the point $(1, 2)$ is,

$$SS_1 = T^2$$

Therefore,

$$(3x^2 + 2y^2 - 5)(3(1)^2 + 2(2)^2 - 5) = (3x(1) + 2y(2) - 5)^2$$

$$(3x^2 + 2y^2 - 5)6 = (3x + 4y - 5)^2$$

$$9x^2 - 4y^2 - 24xy + 30x + 40y - 55 = 0$$

Hence,

$$a = 9$$

$$h = -12$$

$$b = -4$$

The angle between the line is calculated as,

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$= \frac{2\sqrt{144 + 36}}{5}$$

$$= \frac{12\sqrt{5}}{5}$$

$$\theta = \tan^{-1}\left(\frac{12\sqrt{5}}{5}\right)$$

59. The normal to the hyperbola is written as,

$$y = \frac{-l}{m} + \frac{1}{m}$$

So,

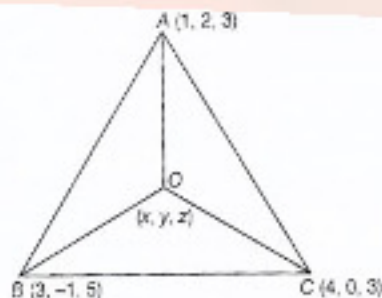
$$\left(\frac{1}{m}\right)^2 = \frac{\left(\frac{-l}{m}\right)^2 (a^2 + b^2)^2}{a^2 - b^2 \left(\frac{-l}{m}\right)^2}$$

$$\frac{1}{m^2} = \frac{\left(\frac{l^2}{m^2}\right)(a^2 + b^2)^2}{\frac{m^2 a^2 - l^2 b^2}{m^2}}$$

$$\frac{1}{m^2} = \frac{l^2 (a^2 + b^2)^2}{m^2 a^2 - l^2 b^2}$$

$$m^2 a^2 - l^2 b^2 = l^2 m^2 (a^2 + b^2)^2$$

60. Consider the figure shown below.



Let $O(x, y, z)$ be the circumcentre of $\triangle ABC$.

$$OA = OB = OC$$

Thus,

$$OA = OB$$

$$OA^2 = OB^2$$

$$(x-1)^2 + (y-2)^2 + (z-3)^2 = (x-3)^2 + (y+1)^2 + (z-5)^2$$

$$4x - 6y + 4z - 21 = 0$$

Similarly,

$$OB^2 = OC^2$$

$$(x-3)^2 + (y+1)^2 + (z-5)^2 = (x-4)^2 + (y-0)^2 + (z+3)^2$$

$$x + y - 8z + 5 = 0$$

Similarly

$$OA^2 = OC^2$$

$$(x-1)^2 + (y-2)^2 + (z-3)^2 = (x-4)^2 + (y-0)^2 + (z+3)^2$$

$$6x - 4y - 12z - 11 = 0$$

On solving the obtained equations,

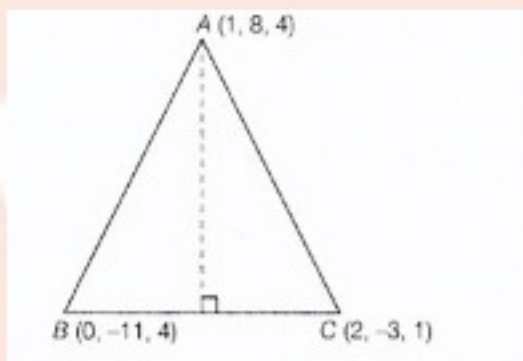
$$x = \frac{7}{2}$$

$$y = -\frac{1}{2}$$

$$z = 1$$

Therefore, the circumcentre of $\triangle ABC$ is $\left(\frac{7}{2}, \frac{-1}{2}, 1\right)$

61. Consider the figure shown below.



The equation of line BC is,

$$\frac{x}{2} = \frac{y+11}{8} = \frac{z-4}{-3} = \lambda$$

The coordinates of any point on the line BC are,

$$(2\lambda, 8\lambda - 11, -3\lambda + 4)$$

Thus, the coordinates of D are,

$$(2\lambda, 8\lambda - 11, -3\lambda + 4)$$

DR's of BC are $2, 8, -3$ and DR's of AD are $2\lambda - 1, 8\lambda - 19, -3\lambda$

Also AD is perpendicular to BC ,

$$\begin{aligned}2(2\lambda - 1) + 8(8\lambda - 19) + (-3)(-3\lambda) &= 0 \\4\lambda - 2 + 64\lambda - 152 + 9\lambda &= 0 \\77\lambda &= 154 \\\lambda &= 2\end{aligned}$$

62. Since, the plane is perpendicular to the given plane.

$$2a + b + 2c = 0$$

And,

$$3a + 3b + 2c = 0$$

By cross multiplication method.

$$\begin{aligned}\frac{a}{2-6} &= \frac{-b}{4-6} = \frac{c}{6-3} \\\frac{a}{-4} &= \frac{b}{2} = \frac{c}{3}\end{aligned}$$

Therefore, the required equation of plane is,

$$\begin{aligned}-4(x-4) + 2(y-4) + 3z &= 0 \\-4x + 16 + 2y - 8 + 3z &= 0 \\4x - 2y - 3z &= 8\end{aligned}$$

63. Consider the limit.

$$y = \lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$$

It is solved as,

$$\begin{aligned}y &= \lim_{x \rightarrow 0} \frac{2 \sin^2 x (3 + \cos x)}{x \tan 4x} \\&= 2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \cdot \lim_{x \rightarrow 0} \frac{x}{\tan 4x} \cdot \lim_{x \rightarrow 0} (3 + \cos x) \\&= 2 \cdot 1 \cdot \frac{1}{4} \cdot 1 \cdot 4 \\&= 2\end{aligned}$$

64. The function is continuous in the interval,

$(-\infty, 1)$, $(1, 3)$, $(3, 5)$ and $(5, \infty)$

Check for continuity at $x = 1$.

$$\begin{aligned}\lim_{x \rightarrow 1^-} f(x) &= 1 - 1 \\&= 0\end{aligned}$$

And,

$$\begin{aligned}\lim_{x \rightarrow 1^+} f(x) &= 2 - 1 \\&= 1\end{aligned}$$

So, the function is not continuous at $x = 1$.

Check for continuity at $x = 3$.

$$\begin{aligned}\lim_{x \rightarrow 3^-} f(x) &= 2 - 9 \\&= -7\end{aligned}$$

And,

$$\begin{aligned}\lim_{x \rightarrow 3^+} f(x) &= 3 - 10 \\ &= -7\end{aligned}$$

Therefore, the function is continuous at $x = 3$

Check for continuity at $x = 5$.

$$\begin{aligned}\lim_{x \rightarrow 5^-} f(x) &= 5 - 10 \\ &= -5\end{aligned}$$

And,

$$\begin{aligned}\lim_{x \rightarrow 5^+} f(x) &= 2 \times 5 \\ &= 10\end{aligned}$$

Thus, the points of discontinuity of f is $\{1, 5\}$.

65. Consider the equation.

$$y = \tan^{-1} x \left\{ \frac{x}{1 + \sqrt{1 - x^2}} \right\} + \sin \left\{ 2 \tan^{-1} \sqrt{\frac{1 - x}{1 + x}} \right\}$$

Substitute $\cos 2\theta$ for x in the above equation.

$$\begin{aligned}
 y &= \tan^{-1} \left\{ \frac{\cos 2\theta}{1 + \sqrt{\sin^2 2\theta}} \right\} + \sin \left\{ 2 \tan^{-1} \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}} \right\} \\
 &= \tan^{-1} \left\{ \frac{\cos 2\theta}{1 + \sin 2\theta} \right\} + \sin \left\{ 2 \tan^{-1} \sqrt{\frac{2 \sin^2 \theta}{2 \cos^2 \theta}} \right\} \\
 &= \tan^{-1} \left\{ \frac{\cos^2 \theta - \sin^2 \theta}{(\cos \theta + \sin \theta)^2} \right\} + \sin \{ 2 \tan^{-1} (\tan \theta) \} \\
 &= \tan^{-1} \left\{ \frac{\cos \theta - \sin \theta}{(\cos \theta + \sin \theta)} \right\} + \sin \{ 2\theta \}
 \end{aligned}$$

Solve further,

$$\begin{aligned}
 y &= \tan^{-1} \left\{ \frac{1 - \tan \theta}{1 + \tan \theta} \right\} + \sin 2\theta \\
 y &= \tan^{-1} \left(\tan \left(\frac{\pi}{4} - \theta \right) \right) + \sin 2\theta \\
 &= \frac{\pi}{4} - \theta + \sin 2\theta \\
 &= \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x + \sqrt{1 - x^2}
 \end{aligned}$$

Differentiate both side with respect to x .

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{2} \cdot \frac{1}{\sqrt{1 - x^2}} + \frac{1}{2\sqrt{1 - x^2}} (-2x) \\
 &= \frac{1 - 2x}{2\sqrt{1 - x^2}}
 \end{aligned}$$

66. Consider the given function.

$$f(x) = \log \left\{ e^x \left(\frac{x-2}{x+2} \right)^{\frac{3}{4}} \right\}$$

It is simplified as,

$$\begin{aligned} f(x) &= \log e^x + \log \left(\frac{x-2}{x+2} \right)^{\frac{3}{4}} \\ &= x \log e + \frac{3}{4} \log \left(\frac{x-2}{x+2} \right) \\ &= x + \frac{3}{4} \log \left(\frac{x-2}{x+2} \right) \end{aligned}$$

Differentiate with respect to x .

$$\begin{aligned} \frac{d}{dx}(f(x)) &= 1 + \frac{3}{4} \left[\frac{x+2}{x-2} \left\{ \frac{(x+2) \cdot 1 - (x-2) \cdot 1}{(x+2)^2} \right\} \right] \\ &= 1 + \frac{3}{4} \left(\frac{4}{x^2 - 4} \right) \\ &= 1 + \frac{3}{x^2 - 4} \end{aligned}$$

$$\left. \frac{d}{dx} f(x) \right|_{x=3} = 1 + \frac{3}{5}$$

Solve further,

$$\left. \frac{d}{dx} f(x) \right|_{x=3} = \frac{8}{5}$$

67. Consider the given equation.

$$y = \frac{\sinh^{-1} x}{\sqrt{1+x^2}}$$

$$\sqrt{1+x^2} y = \sinh^{-1} x$$

Differentiate the above equation with respect to x .

$$\sqrt{1+x^2} y_1 + y \frac{1}{2\sqrt{1+x^2}} \cdot 2x = \frac{1}{\sqrt{1+x^2}}$$

$$(1+x^2) y_1 + xy = 1$$

Differentiate again with respect to x

$$(1+x^2) y_2 + y_1 2x + xy_1 + y = 0$$

$$(1+x^2) y_2 + 3xy_1 + y = 0$$

68. Consider the given equation of curves.

$$x^2 = 3y$$

And,

$$x^2 + y^2 = 4$$

Substitute $3y$ for x^2 in the above equation.

$$y^2 + 3y - 4 = 0$$

$$y(y+4) - 1(y+4) = 0$$

$$(y+4)(y-1) = 0$$

$$y = -4, 1$$

If $y = -4$,

$$x^2 = -12$$

This is not possible.

$$x = \pm\sqrt{3}$$

The point of intersection are $(\sqrt{3}, 1)$ and $(-\sqrt{3}, 1)$.

So,

$$\frac{dy}{dx} = \frac{2x}{3}$$

$$m_1 = \frac{2}{3}(\sqrt{3})$$

Also,

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$m_2 = -\frac{x}{y}$$

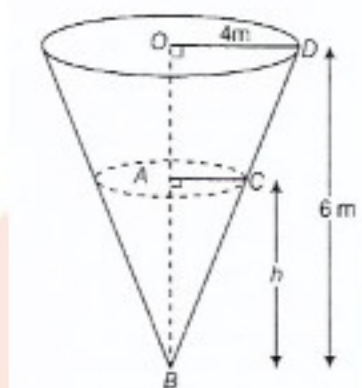
$$= \frac{\sqrt{3}}{1}$$

$$= \sqrt{3}$$

The angle between the curve is calculated as,

$$\begin{aligned}\tan \theta &= \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| \\&= \left| \frac{-\sqrt{3} - \frac{2\sqrt{3}}{3}}{1 + \left(\frac{2\sqrt{3}}{3}\right)(-\sqrt{3})} \right| \\&= \frac{5}{\sqrt{3}} \\ \theta &= \tan^{-1} \left(\frac{5}{\sqrt{3}} \right)\end{aligned}$$

69. Consider the figure shown below.



It is given that,

$$\frac{dv}{dt} = 3 \text{ m}^3/\text{min}$$

Now, $\Delta BOD \sim \Delta BAC$

Hence,

$$\frac{BO}{BA} = \frac{OD}{AC}$$

$$\frac{6}{h} = \frac{4}{r}$$

$$r = \frac{2}{3}h$$

Therefore,

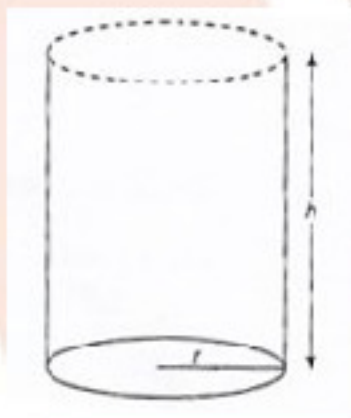
$$V = \frac{1}{3}\pi \left(\frac{2}{3}h\right)^2 h$$

$$= \frac{4}{27}\pi h^3$$

So,

$$\begin{aligned}\frac{dV}{dt} &= \frac{12}{27} \pi h^2 \frac{dh}{dt} \\ \left. \frac{dh}{dt} \right|_{h=3} &= \frac{27 \times 3}{12\pi(9)} \\ &= \frac{3}{4\pi} \text{ m/min}\end{aligned}$$

70. Consider the figure shown below.



The volume of the cylinder is given as,

$$V = \pi r^2 h$$

The area of the metal sheet used to form the vessel is,

$$\begin{aligned} S &= 2\pi rh + \pi r^2 \\ &= 2\pi r \frac{V}{\pi r^2} + \pi r^2 \\ S(r) &= \frac{2V}{r} + \pi r^2 \end{aligned}$$

For minimum value of $S(r)$ put $S'(r) = 0$.

$$\begin{aligned} -\frac{2V}{r^2} + 2\pi r &= 0 \\ \frac{2V}{r^2} &= 2\pi r \\ r^3 &= \frac{V}{\pi} \\ r &= \sqrt[3]{\frac{V}{\pi}} \end{aligned}$$

Thus,

$$\begin{aligned} V &= \pi \left(\frac{V}{\pi} \right)^{\frac{2}{3}} h \\ h &= \frac{V^{1/3}}{\pi^{1/3}} \\ h &= \sqrt[3]{\frac{V}{\pi}} \end{aligned}$$

71. Consider the function.

$$f(x) = (x-1)(x-2)$$

So,

$$f(x) = x^2 - 3x + 2$$

According to the LMV theorem,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$2c - 3 = \frac{\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right) - (-1)(-2)}{\frac{1}{2}}$$

$$2c - 3 = -\frac{5}{2}$$

$$4c - 6 = -5$$

Solve further,

$$4c - 6 = -5$$

$$c = \frac{1}{4} \in \left(0, \frac{1}{2}\right)$$

72. Consider the integral.

$$I = \int \frac{x^{e-1} + e^{x-1}}{x^e + e^x} dx$$

Let,

$$x^e + e^x = t$$

$$(ex^{e-1} + e^x)dx = dt$$

$$(x^{e-1} + e^{x-1}) = \frac{dt}{e}$$

Therefore,

$$I = \frac{1}{e} \int \frac{dt}{t}$$

$$= \frac{1}{e} \log|t| + c$$

$$= \frac{1}{e} \log|x^e + e^x| + c$$

73. Consider the integral.

$$I = \int \frac{x + \sin x}{1 + \cos x} dx$$

It is solved as,

$$\begin{aligned} I &= \int \frac{x + \sin x}{2 \cos^2 x/2} dx \\ &= \frac{1}{2} \int x \sec^2 x/2 dx + \int \frac{2 \sin x/2 \cos x/2}{2 \cos^2 x/2} dx \\ &= \frac{1}{2} \int x \sec^2 x/2 dx + \int \tan \frac{x}{2} dx \\ &= \frac{1}{2} \left[x \frac{\tan x/2}{1/2} - \int \tan \frac{x}{2} dx \right] + \int \tan \frac{x}{2} dx \end{aligned}$$

And,

$$\begin{aligned} I &= x \tan \frac{x}{2} - \int \tan \frac{x}{2} dx + \int \tan \frac{x}{2} dx \\ &= x \tan \frac{x}{2} + C \end{aligned}$$

74. Consider the integral.

$$I = \int \frac{5x^2 + 3}{x^2(x^2 - 2)} dx$$

Let $x^2 = y$

$$\frac{5x^2 + 3}{x^2(x^2 - 2)} = \frac{5y + 3}{y(y - 2)}$$

Thus,

$$\frac{5y+3}{y(y-2)} = \frac{A}{y} + \frac{B}{y-2}$$

$$5y+3 = A(y-2) + By$$

Compare the coefficient of like terms.

$$A + B = 5$$

$$2A = -3$$

So,

$$A = \frac{-3}{2}$$

$$B = \frac{13}{2}$$

Hence,

$$\frac{5x^2+3}{x^2(x^2-2)} = -\frac{3}{2} \cdot \frac{1}{x^2} + \frac{13}{2} \cdot \frac{1}{x^2-2}$$

Therefore,

$$\begin{aligned} I &= \int -\frac{3}{2} \cdot \frac{1}{x^2} + \frac{13}{2} \cdot \frac{1}{x^2-2} dx \\ &= -\frac{3}{2} \left(-\frac{1}{x} \right) + \frac{13}{2} \left(\frac{1}{2\sqrt{2}} \right) \log \left| \frac{x-\sqrt{2}}{x+\sqrt{2}} \right| + c \\ &= \frac{3}{2x} + \frac{13}{4\sqrt{2}} \log \left| \frac{x-\sqrt{2}}{x+\sqrt{2}} \right| + c \end{aligned}$$

75. Consider the integral.

$$I_n = \int \frac{\sin nx}{\cos x} dx$$

It is solved as,

$$\begin{aligned} I_n &= \int \frac{\sin(nx - x + x)}{\cos x} dx \\ &= \int \frac{\sin((n-1)x + x)}{\cos x} dx \\ &= \int \frac{\sin(n-1)x \cos x + \cos(n-1)x \sin x}{\cos x} dx \\ &= \int \sin(n-1) dx + \int \frac{\cos(n-1)x \sin x}{\cos x} dx \end{aligned}$$

Solve further,

$$\begin{aligned} I &= \int \sin(n-1) dx + \frac{1}{2} \int \frac{2 \sin x \cos(n-1)x}{\cos x} dx \\ &= -\frac{\cos(n-1)}{n-1} + \frac{1}{2} \int \frac{\sin nx + \sin(2-n)x}{\cos x} dx \\ &= -\frac{\cos(n-1)}{n-1} + \frac{1}{2} \int \frac{\sin nx}{\cos x} dx - \frac{1}{2} \int \frac{\sin(n-2)x}{\cos x} dx \\ &= -\frac{\cos(n-1)}{n-1} + \frac{1}{2} I_n - \frac{1}{2} I_{n-2} \end{aligned}$$

Solve further,

$$\left(1 - \frac{1}{2}\right)I_n = -\frac{\cos(n-1)}{n-1} - \frac{1}{2}I_{n-2}$$

$$I_n = \frac{-2}{n-1} \cos(n-1)x - I_{n-2}$$

76. Consider the limit.

$$y = \lim_{n \rightarrow \infty} \left[\frac{1^k + 2^k + 3^k + \dots + n^k}{n^{k+1}} \right]$$

It is solved as,

$$\begin{aligned} y &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\left(\frac{1}{n}\right)^k + \left(\frac{2}{n}\right)^k + \dots + \left(\frac{n-1}{n}\right)^k + 1 \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \left(\frac{r}{n}\right)^k + \lim_{n \rightarrow \infty} \frac{1}{n} \\ &= \int_0^1 x^k dx + 0 \\ &= \frac{1}{k+1} \end{aligned}$$

77. Consider the integral.

$$I = \int_0^1 f(k-1+x) dx$$

Let,

$$k-1+x=t$$

$$dx=dt$$

Therefore,

$$I = \int_{k-1}^k f(t) dt$$

$$= \int_{k-1}^k f(x) dx$$

Now,

$$\begin{aligned} \sum_{k=1}^{10} \int_0^1 f(k-1+x) dx &= \sum_{k=1}^{10} \int_{k-1}^k f(x) dx \\ &= \int_0^1 f(x) dx + \int_1^2 f(x) dx + \cdots + \int_9^{10} f(x) dx \\ &= \int_0^{10} f(x) dx \\ &= 5 \end{aligned}$$

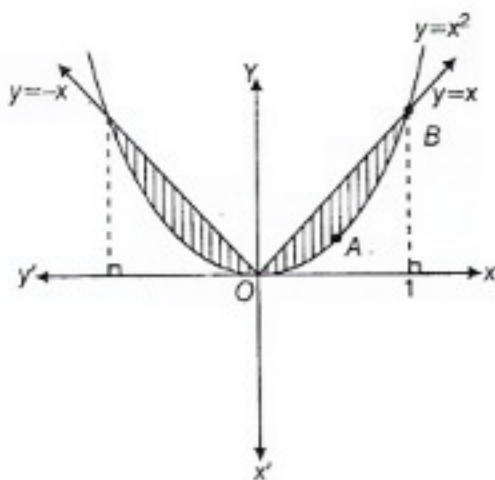
78. Consider the given equation.

$$y = x^2$$

And

$$\begin{aligned} y &= |x| \\ &= \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \end{aligned}$$

Consider the graph shown below.



Substitute x for y in the above equation.

$$x = x^2$$

$$x(x-1) = 0$$

$$x = 0, 1$$

Thus, the required area is,

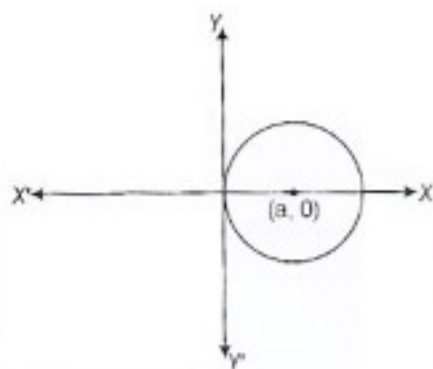
$$\text{Area} = 2 \int_0^1 x - x^2 dx$$

$$= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= 2 \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$= \frac{1}{3} \text{ sq units}$$

79. Consider the figure shown below.



So,

$$(x - a)^2 + y^2 = a^2$$

$$x^2 + a^2 - 2ax + y^2 = a^2$$

$$x^2 + y^2 = 2ax$$

Differentiate with respect to x

$$2x + 2y \frac{dy}{dx} = 2a$$

Hence,

$$x^2 + y^2 = \left(2x + 2y \frac{dy}{dx} \right) x$$

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

80. Consider the differential equation.

$$(x - 4y^3) \frac{dy}{dx} - y = 0$$

It is solved as,

$$\frac{dx}{dy} = \frac{(x - 4y^3)}{y}$$

$$\frac{dx}{dy} - \frac{x}{y} = -4y^2$$

Calculate the integrating factor.

$$\begin{aligned} IF &= e^{\int \frac{-1}{y} dy} = e^{-\log y} \\ &= e^{\log y^{-1}} \\ &= \frac{1}{y} \end{aligned}$$

The solution is,

$$x \left(\frac{1}{y} \right) = \int \frac{1}{y} (-4y^2) dy + c$$

$$\frac{x}{y} = -4 \frac{y^2}{2} + c$$

$$x = -2y^3 + cy$$

$$x + 2y^3 = cy$$

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Solutions – Physics

81. The formula for the internal resistance of the cell is,

$$r = R \left(\frac{l_1}{l_2} - 1 \right)$$

The percentage change in the internal resistance is,

$$\begin{aligned} \frac{\Delta r}{r} &= \frac{\Delta R}{R} \times 100 + \frac{\Delta \left(\frac{l_1}{l_2} \right)}{\left(\frac{l_1}{l_2} \right)} \times 100 \\ &= 4 + 0.2 \\ &= 4.2\% \end{aligned}$$

82. The total time required by the two cars to meet,

$$\begin{aligned} t &= \frac{36}{54 + 36} \\ t &= 0.4 \text{ h} \end{aligned}$$

The distance covered by the bird before the cars meet each other is,

$$\begin{aligned}d &= 36 \times 0.4 \\&= 14.4 \text{ km or} \\&= 14400 \text{ m}\end{aligned}$$

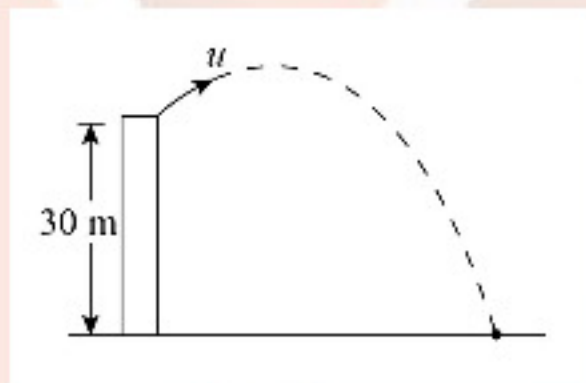
83. Given Data:

$$u = 5 \text{ m/s}$$

$$a = -10 \text{ m/s}^2$$

$$h = -30 \text{ m}$$

The figure below represents the projectile motion.



$$h = ut + \frac{1}{2}at^2$$

$$-30 = 5t - \frac{1}{2} \times 10 \times t^2$$

By solving the above equation.

$$t = 3 \text{ or } t \neq -2$$

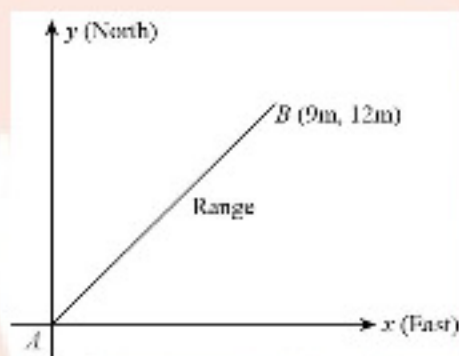
$$t = 3 \text{ s}$$

The distance covered in east direction is,

$$d_x = 3 \times 3 = 9 \text{ m}$$

The distance covered in north direction is,

$$d_y = 4 \times 3 = 12 \text{ m}$$



The range is,

$$R = \sqrt{9^2 + 12^2}$$

$$R = \sqrt{225} \text{ m}$$

$$R = 15 \text{ m}$$

84. The equation of the projectile is given as,

$$y = Px - Qx^2 \quad \dots\dots (I)$$

The distance travelled,

$$y = x \tan \theta - \frac{1}{2} g \left(\frac{x^2}{u^2 \cos^2 \theta} \right) \quad \dots\dots (II)$$

From equation (I) and (II),

$$P = \tan \theta$$

$$Q = \frac{g}{2u^2 \cos^2 \theta}$$

The formula for maximum height and range is,

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

$$R = \frac{u^2 \sin 2\theta}{2g}$$

By dividing the two equations,

$$\frac{H}{R} = \frac{\sin^2 \theta}{2 \times \sin 2\theta}$$

$$\frac{H}{R} = \frac{\sin^2 \theta}{2 \times 2 \sin \theta \cos \theta}$$

$$\frac{H}{R} = \frac{\tan \theta}{4}$$

$$\frac{H}{R} = \frac{P}{4}$$

85. The formula for the resultant acceleration when the lift goes down,

$$a' = (g - a)$$

The force exerted by the man on the floor of the lift is,

$$F = ma'$$

$$F = m(g - a)$$

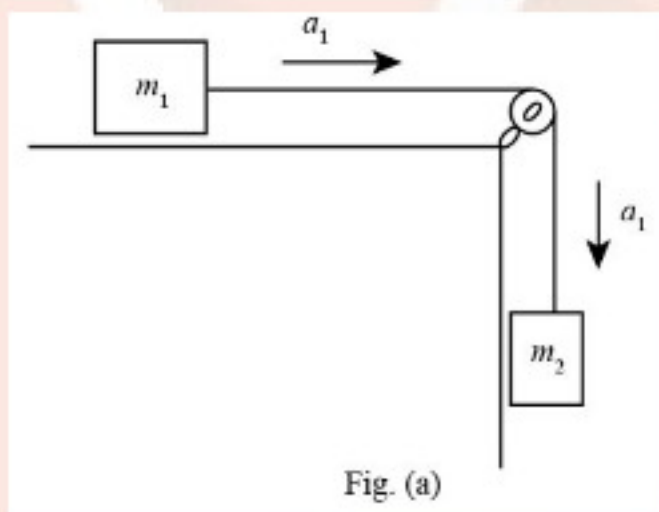
$$150 = 60(10 - a)$$

$$a = 7.5 \text{ m/s}^2$$

86. Given Data:

$$m_1 = 3 \text{ kg}, m_2 = 4 \text{ kg}$$

Consider the diagram as shown below,



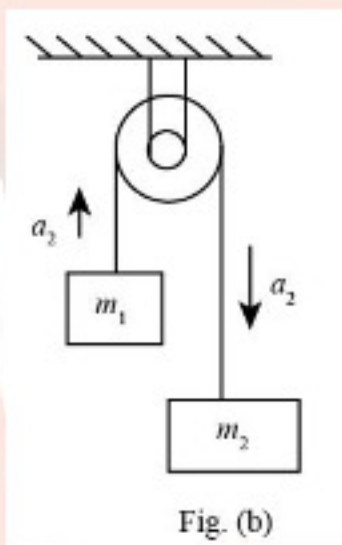
Balancing the force,

$$(m_1 + m_2)a_1 = m_2g$$

$$a_1 = \frac{m_2g}{m_1 + m_2}$$

$$a_1 = \frac{4 \times 10}{3 + 4} = \frac{40}{7} \text{ m/s}^2$$

Consider the diagram as shown below,



By balancing the forces,

$$(m_1 + m_2)a = (m_2 - m_1)g$$

$$a = \frac{m_2 - m_1}{(m_1 + m_2)}g$$

$$a = \frac{4 - 3}{4 + 3} \times 10$$

$$a = \frac{10}{7} \text{ m/s}^2$$

87. The work done to displace the girl can be calculated as,

$$W = mg \sin 60^\circ ds$$

$$W = mg \sin 60^\circ (v dt)$$

The power generated at that instant is,

$$P = mg \sin 60^\circ (v)$$

$$P = 50(9.8) \left(\frac{\sqrt{3}}{2} \right) (2)$$

$$P = 490\sqrt{3} \text{ W}$$

88. The acceleration of the block can be calculated as,

$$F = (m + M)a$$

$$a = \frac{F}{m + M}$$

Since there is no relative sliding, the frictional force on upper block is,

$$f = ma$$

$$f = m \left(\frac{F}{M + m} \right)$$

The distance covered can be calculated as,

$$D = \frac{1}{2}at^2$$

$$D = \frac{1}{2} \left(\frac{F}{M+m} \right) t^2$$

The work done by the frictional force is,

$$W = f \times s$$

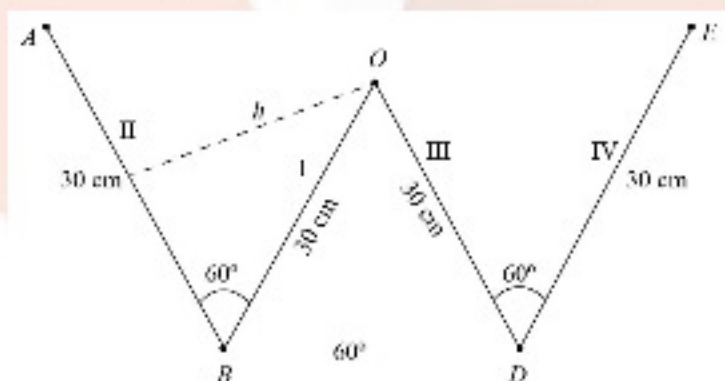
$$= \frac{mF}{M+m} \times \frac{1}{2} \frac{Ft^2}{m+m}$$

$$= \frac{mF^2t^2}{2(m+M)}$$

89. The mass of each rod is, $m = 400 \text{ g} = 0.4 \text{ kg}$

The length of each rod is, $l = 30 \text{ cm} = 0.3 \text{ m}$

The figure below represents the uniform rods,



$$h^2 = l^2 - \left(\frac{l}{2}\right)^2$$

$$h^2 = \frac{3}{4}l^2$$

The moment of inertia of I & III is same and II and IV is same.

The moment of inertia of the system,

$$MI = 2 \times (MI \text{ of } I) + 2 \times (MI \text{ of } II)$$

$$= \frac{ml^2}{3} \times 2 + 2 \left(\frac{ml^2}{12} + mh^2 \right)$$

$$= \frac{2}{3}ml^2 + 2ml^2 \left(\frac{10}{12} \right)$$

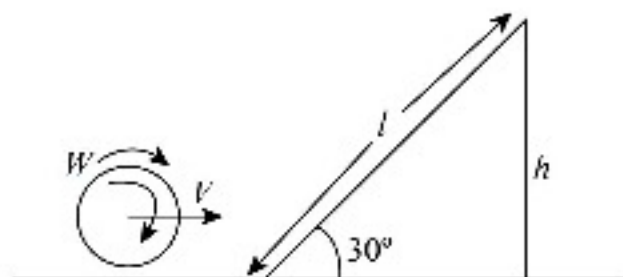
$$= \frac{7}{3}ml^2$$

Substitute the values in the above equation.

$$\begin{aligned} MI &= \frac{7}{3} \times 0.4 \times 0.3 \times 0.3 \\ &= 0.084 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

90. The velocity of the solid sphere is $v = 4 \text{ m/s}$

The figure below represents the solid sphere moves along the inclined plane.



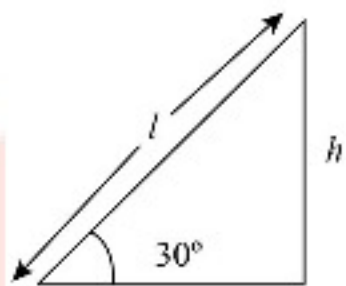
The kinetic energy at the bottom, is equal to the sum of kinetic energy due to translation and rotation,

$$\begin{aligned}
 KE &= KE_t + KE_r \\
 &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\
 &= \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right)\left(\frac{v^2}{r^2}\right) \\
 &= \frac{1}{2}\left(\frac{7}{5}mv^2\right)
 \end{aligned}$$

From the conservation of energy, at height h ,

$$\begin{aligned}
 KE &= mgh \\
 \frac{1}{2}\left(\frac{7}{5}mv^2\right) &= mgh \\
 \frac{1}{2}\left(\frac{7}{5}(4)^2\right) &= (10)h \\
 h &= 1.12 \text{ m}
 \end{aligned}$$

The maximum distance travelled is,



$$l = \frac{h}{\sin 30^\circ}$$

$$l = \frac{1.12}{\sin 30^\circ}$$

$$l = 22.4 \text{ m}$$

91. The formula for the time period of the pendulum,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T \propto \sqrt{\frac{L}{g}}$$

The ratio of T_1 / T_2 is

$$\frac{T_1}{T_2} = \frac{\sqrt{L/g}}{\sqrt{2L/3g}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{3}{2}}$$

92. The position where the gravitation field is zero.

$$\frac{G(4m)}{x} = \frac{G(9m)}{(r-x)}$$

Here, x is distance is from 4 m mass.

$$\frac{4}{x} = \frac{9}{(r-x)}$$

$$4(r-x) = 9x$$

$$4r = 13x$$

$$x = \frac{4}{13}r$$

The point P is at a distance $\frac{4}{13}r$ from mass 4 m and

$$\left(h - \frac{4}{13}r\right) = \frac{9r}{13} \text{ from mass } m.$$

$$v = -\frac{G(4m)}{\left(\frac{4}{13}r\right)} - \frac{G(9m)}{\frac{9r}{13}}$$

$$v = -\frac{26Gm}{r}$$

93. Given:

$$\text{Length: } \frac{L_s}{L_b} = a$$

$$\text{Area of cross section: } \frac{A_s}{A_b} = b$$

$$\text{Young's modules: } \frac{E_s}{E_b} = b$$

The formula for the increase in the length,

$$\frac{\Delta L}{L} = \frac{\sigma}{E}$$

$$\Delta L = \frac{F / A}{E} L$$

$$\Delta L = \frac{FL}{AE}$$

The ratio of change in length of steel and brass,

$$\frac{\Delta L_s}{\Delta L_b} = \left(\frac{F_s L_s}{A_s E_s} \right) \left(\frac{A_s E_s}{F_s L_s} \right)$$

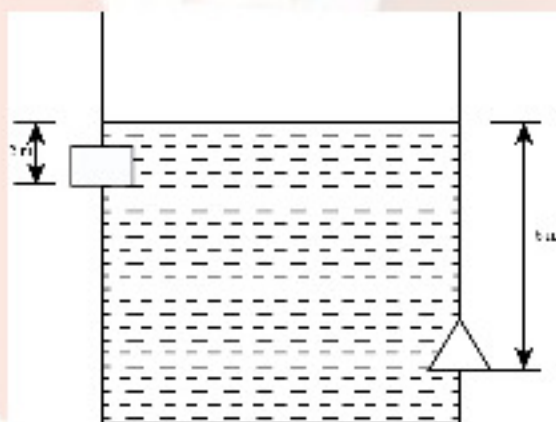
$$\frac{\Delta L_s}{\Delta L_b} = \left(\frac{F_s}{F_b} \right) \left(\frac{L_s}{L_b} \right) \left(\frac{A_b}{A_s} \right) \left(\frac{E_b}{E_s} \right)$$

$$\frac{\Delta L_s}{\Delta L_b} = \frac{7}{4} (a) \left(\frac{1}{b} \right) \left(\frac{1}{c} \right)$$

$$\frac{\Delta L_s}{\Delta L_b} = \frac{7a}{4bc}$$

$$\frac{\Delta L_b}{\Delta L_s} = \frac{4bc}{7a}$$

94. The figure below represents the tanks with the two holes,



The formula for the velocity is,

$$v = \sqrt{2gh}$$

The volumetric flow rate of square hole is,

$$\begin{aligned}
 V_s &= Av \\
 &= x^2 \times \sqrt{2g \times 2}
 \end{aligned}$$

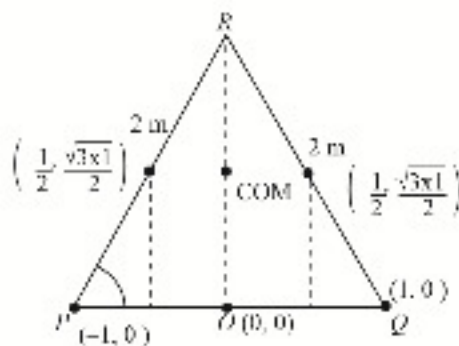
The volumetric flow rate of triangular hole is,

$$\begin{aligned}
 V_t &= Av \\
 &= \frac{\sqrt{3}}{4} \times 4^2 \times \sqrt{2g \times 6}
 \end{aligned}$$

Since the flow rate is same,

$$\begin{aligned}
 V_s &= V_t \\
 x^2 \times \sqrt{2g \times 2} &= \frac{\sqrt{3}}{4} \times 4^2 \times \sqrt{2g \times 6} \\
 x^2 &= \sqrt{3} \times \sqrt{3} \\
 x &= \sqrt{3} = 1.732 \text{ cm}
 \end{aligned}$$

95. The figure below represents the three uniform thin rods in the form of triangle.



The centre of mass of the system,

$$r_{COM} = \frac{m_1 r_1 + m_2 r_2 + m_3 r_3}{(m_1 + m_2 + m_3)}$$

$$(r_{COM})_i = \frac{1}{2} \left(\frac{\sqrt{3}}{2} \times 1 \times 2 \right) \hat{j}$$

$$(r_{COM})_i = \frac{1}{\sqrt{3}} \hat{j}$$

The new length of the rod due to expansion is,

$$l_f = l_i (1 + \alpha \Delta T)$$

$$l_f = l_i \left(1 + \frac{\gamma}{3} \Delta T \right)$$

Substitute the values in the above equation.

$$l_f = 2 \left(1 + \frac{12\sqrt{3} \times 10^{-6}}{3} \times 50 \right)$$

$$l_f = 2 \left(1 + 4 \times 50 \times \sqrt{3} \times 10^{-6} \right)$$

$$l_f = 2 \left(1 + 2\sqrt{3} \times 10^{-4} \right)$$

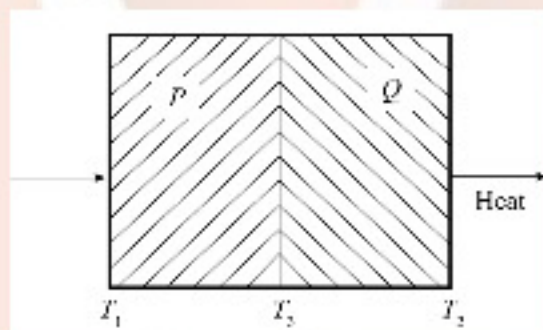
The final centre of mass can be calculated as,

$$\begin{aligned} (r_{COM})_f &= \frac{1}{3} \left(\frac{\sqrt{3}}{2} \times (1 + 2\sqrt{3} \times 10^{-4}) \times 2 \right) \hat{j} \\ &= \left(\frac{1}{\sqrt{3}} + 2 \times 10^{-4} \right) \hat{j} \end{aligned}$$

The increase in y coordinate of the centre of mass of the system of the rods is,

$$\begin{aligned}\delta y &= (r_{COM})_f - (r_{COM})_i \\ &= \left(\frac{1}{\sqrt{3}} + 2 \times 10^{-4} - \frac{1}{\sqrt{3}} \right) \hat{j} \\ &= 2 \times 10^{-4} \text{ m}\end{aligned}$$

96. The figure below represents the wall P and Q.



It is given that,

$$k_Q = \frac{k_P}{2}$$

$$k_P = 2k_Q$$

And

$$T_1 - T_2 = 24^\circ\text{C}$$

In steady state, the rate of flow heat through layer P and Q is same,

$$\frac{k_P a (T_1 - T_3)}{x} = \frac{k_Q a (T_3 - T_2)}{x}$$

Substitute the values in the above equation.

$$\frac{2k_Q a (T_1 - T_3)}{x} = \frac{k_Q a (T_3 - T_2)}{x}$$

$$2(T_1 - T_3) = (T_3 - T_2)$$

$$2T_1 - 3T_3 + T_2 = 0$$

From equation (I),

$$2T_1 - 3T_3 + T_1 - 24 = 0$$

$$3T_1 - 3T_3 = 24$$

$$T_1 - T_3 = 8^\circ\text{C}$$

97. The relation is given as,

$$V = KT^2$$

From ideal gas equation,

$$PV = RT$$

$$PV = R \left(\frac{V}{K} \right)^{1/2}$$

$$PV^{1/2} = a \text{ const}$$

From the above equation, $\gamma = 1/2$.

The work done by the gas is,

$$W = \frac{nR(T_i - T_f)}{1 - \gamma}$$

$$W = \frac{R(60)}{1 - 1/2}$$

$$W = 120R$$

98. The total work done is equal to the area under the cyclic process.

$$W = (3P - P)(2V - V)$$

$$W = 2PV$$

The total heat supplied to the system,

$$\begin{aligned}
 Q &= Q_{AB} + Q_{BC} \\
 &= nc_v \Delta T + nc_p \Delta T \\
 &= \frac{3}{2} R \Delta T + \frac{5}{2} R \Delta T \\
 &= \frac{3}{2} V \Delta P + \frac{5}{2} 3P \Delta V \\
 &= \frac{3}{2} V (3P - 2P) + \frac{5}{2} 3P (2V - V) \\
 &= 3PV + \frac{15}{2} PV \\
 &= \frac{21}{2} PV
 \end{aligned}$$

The efficiency is calculated as,

$$\begin{aligned}
 \eta &= \frac{W}{Q} \times 100 \\
 &= \frac{2PV}{\frac{21}{2} PV} \times 100 \\
 &= 19.04\%
 \end{aligned}$$

Thus, no option is correct.

99. The translation kinetic energy of a molecule of gas is $= \frac{3}{2} kT$

The formula for the kinetic energy is,

$$\frac{3}{2}kT = eV$$

$$\frac{3}{2}(1.38 \times 10^{-23})T = (1.6 \times 10^{-19})(10)$$

$$T = 7.73 \times 10^4$$

$$T = 77.3 \times 10^3 \text{ K}$$

100. The equation of the transverse displacement is,

$$y = 0.03 \sin\left(\frac{2\pi x}{3}\right) \cos 60\pi t \quad \dots\dots (I)$$

The mass per unit length is given as, $\mu = 0.01 \text{ kg/m}$

The general equation of transverse displacement is,

$$y = a \sin kx \cos \omega t$$

Comparing the above equation with equation (I),

$$\omega = 60\pi, k = \frac{2\pi}{3}$$

The formula for the velocity is,

$$v = \sqrt{\frac{T}{\mu}} = \frac{\omega}{k}$$

Substitute the values in the above equation.

$$\sqrt{\frac{T}{0.01}} = \frac{60\pi}{2\pi/3}$$

$$\sqrt{\frac{T}{0.01}} = 90$$

$$T = 90^2 \times 0.01$$

$$T = 81 \text{ N}$$

101. For the closed organ pipe,

$$f = f_2 = 3f_1$$

Now,

$$3f_1 = \frac{3v}{4L} = \frac{3}{4L} \sqrt{\frac{\gamma p}{\rho_1}}$$

For the open organ pipe,

$$f = f_2 = 2f_1$$

Now,

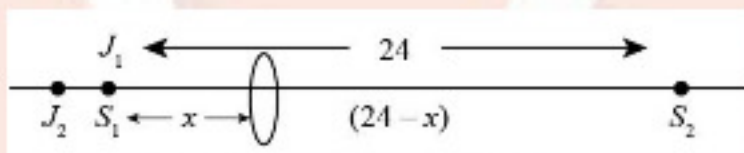
$$2f_1 = \frac{2v}{2L'} = \frac{1}{L'} \sqrt{\frac{\gamma p}{\rho_2}}$$

From same sound frequency,

$$\frac{3}{4L} \sqrt{\frac{\gamma p}{\rho_1}} = \frac{1}{L'} \sqrt{\frac{\gamma p}{\rho_2}}$$

$$L' = \frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$$

102. The figure below represents the convex lens and two point sources,



The given condition satisfies when S_1 is placed in one side such that $u < f$ and the S_2 is placed on the other side of the lens such that $u > f$.

S_1 is the object for lens, from lens formula,

$$\frac{1}{f} = \frac{1}{-y} - \frac{1}{-x}$$

$$\frac{1}{y} = \frac{1}{x} - \frac{1}{-f} \quad \text{..... (I)}$$

S_2 is the object for the lens, then

$$\frac{1}{f} = \frac{1}{y} - \frac{1}{-(24-x)}$$

$$\frac{1}{-y} = \frac{1}{f} - \frac{1}{24-x} \quad \text{..... (II)}$$

From equation (I) and (II),

$$\frac{1}{x} - \frac{1}{f} = \frac{1}{f} - \frac{1}{24-x}$$

$$\frac{1}{x} - \frac{1}{9} = \frac{1}{9} - \frac{1}{24-x}$$

$$\frac{1}{x} + \frac{1}{24-x} = \frac{2}{9}$$

$$x^2 - 24x + 108 = 0$$

By solving the above equation,

$$x = 6 \text{ cm or } 18 \text{ cm}$$

103. The formula for the fringe width is,

$$\beta = \frac{\lambda d}{D}$$

Substitute the given values in the above equation.

$$\beta = \frac{(120 \times 10^{-2})(0.5 \times 10^{-9})}{500 \times 10^{-9}}$$

$$\beta = 12 \times 10^{-4} \text{ m}$$

The number of fringes in 3 mm distance on screen is,

$$n = \frac{3 \text{ mm}}{12 \times 10^{-4}}$$

$$n = \frac{3 \times 10^{-3}}{12 \times 10^{-4}} = 25$$

That is third fringe.

The phase difference is,

$$\phi = 2\pi + 2\pi + \pi$$

$$\phi = 5\pi$$

104. The acceleration of projected charge in y-direction is,

$$a_y = -\frac{F}{m}$$

$$a_y = -\frac{qE}{m}$$

Substitute the given values in the above equation.

$$a_y = -\frac{(1 \times 10^{-6})(1 \times 10^3)}{2 \times 10^{-3}}$$

$$a_y = -\frac{1}{2} \text{ m/s}^2$$

The vertical component of the projection velocity is,

$$u_y = u \sin 45^\circ$$

If the particle is not to hit upper plate, then

$$(u_y)^2 \leq 2a_y(y)$$

$$(u \sin 45^\circ)^2 \leq 2 \times \frac{1}{2} \times 2 \times 10^{-2}$$

$$u^2 \leq 4 \times 10^{-2}$$

$$u \leq 2 \times 10^{-1}$$

$$u \leq 0.2 \text{ m/s}$$

105. The formula for the intensity of the electric field due to infinitely long wire at distance r is,

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$E = \frac{2\lambda}{4\pi\epsilon_0 r}$$

Substitute the given values in the above equation.

$$E = \frac{(9 \times 10^9) 2 \left(\frac{1}{3}\right)}{3 \times 18 \times 10^{-2}}$$

$$E = \frac{1}{3} \times 10^{11} \text{ N/C}$$

The force on the charge is,

$$\begin{aligned}F &= qE \\&= 3 \times 10^{-6} \times \frac{1}{3} \times 10^{11} \\&= 10^5 \text{ N}\end{aligned}$$

106. The voltage across capacitance is,

$$\begin{aligned}V_{cap} &= \frac{Q}{C} \\&= \frac{4 \times 10^{-6}}{2 \times 10^{-6}} \\&= 2 \text{ V}\end{aligned}$$

The across 2Ω resistance is same as of V_{Cap}

Therefore,

$$\begin{aligned}I &= \frac{V}{R} \\I &= \frac{2}{2} = 1 \text{ A}\end{aligned}$$

The formula for the emf of the cell,

$$E = V + Ir$$

Substitute the values in the above equation.

$$\begin{aligned}E &= 2 + 1 \times 0.5 \\&= 2.5 \text{ V}\end{aligned}$$

107. The diameter of the sphere is, $d = 18 \text{ cm} = 18 \times 10^{-2} \text{ m}$

Potential: $V = 25 \text{ kV} = 25 \times 10^3 \text{ V}$

The formula for the charge on the metal sphere is,

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} \right)$$

$$q = \frac{4\pi\epsilon_0(r)V}{1}$$

Substitute the values in the above equation.

$$q = \frac{25 \times 10^3 \times \frac{18}{2} \times 10^{-2}}{9 \times 10^9}$$

$$q = 0.25 \times 10^{-6}$$

$$q = 0.25 \mu\text{C}$$

108. Given data:

$$V = 1.6 \text{ V}$$

$$R = 4 \Omega$$

The current in the wire is,

$$I_1 = \frac{1.6}{4} = 0.4 \text{ A}$$

The emf of the cell is,

$$E = 1.6 + 0.4r \quad \text{..... (I)}$$

When a $4\ \Omega$ resistance is connected in parallel,

$$\frac{1}{R'} = \frac{1}{4} + \frac{1}{4}$$

$$\frac{1}{R'} = \frac{2}{4}$$

$$R' = 2\ \Omega$$

And

$$I_2 = \frac{1.33}{2} = 0.66\ \text{A}$$

$$E = 1.33 + 0.66r \quad \text{..... (II)}$$

From equation (I) and (II),

$$1.6 + 0.4r = 1.33 + 0.66r$$

$$1.6 - 1.33 = 0.66r - 0.4r$$

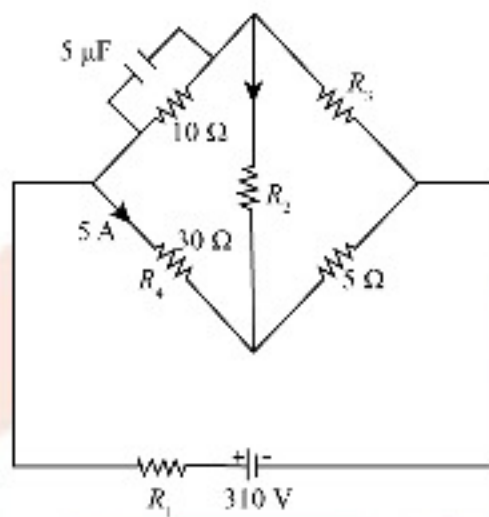
$$r = \frac{0.27}{0.26} \approx 1$$

Substitute the value in equation (I).

$$E = 1.6 + 0.4 \times 1$$

$$E = 2\ \text{V}$$

109. The figure below represents the given circuit.



$$V_{cap} = \frac{Q}{C}$$

$$V_{cap} = \frac{1 \times 10^{-3}}{5 \times 10^{-6}}$$

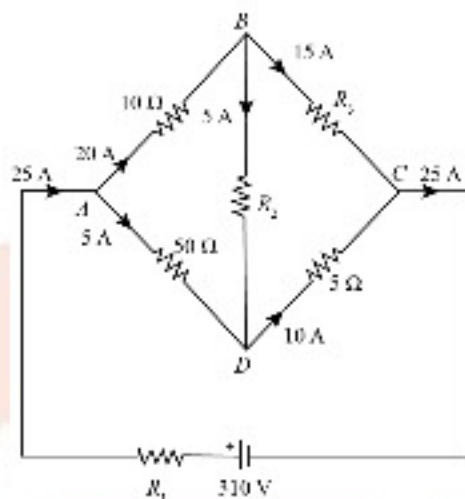
$$V_{cap} = 200\ \text{V}$$

The current across R_5 resistance,

$$I = \frac{V}{R}$$

$$I = \frac{200}{10} = 20\ \text{A}$$

The figure below represents the current distribution.



In loop $BCDB$,

$$-15R_3 + 10 \times 5 + 5R_2 = 0 \quad \text{..... (I)}$$

In loop $ABCD$,

$$-200 - 15R_3 + 50 + 250 = 0$$

$$15R_3 = 100$$

$$R_3 = \frac{100}{15} \Omega$$

From equation (I),

$$5R_2 = 50$$

$$R_2 = 10 \Omega$$

In loop $ADCA$,

$$-250 - 50 + 310 - 25R_1 = 0$$

$$10 = 25R_1$$

$$R_1 = \frac{2}{5} \Omega$$

The required value is,

$$\begin{aligned} \frac{R_1 \times R_2}{R_3} &= \frac{\frac{2}{5} \times 10}{\left(\frac{100}{15}\right)} \\ &= \frac{3}{5} = 0.6 \end{aligned}$$

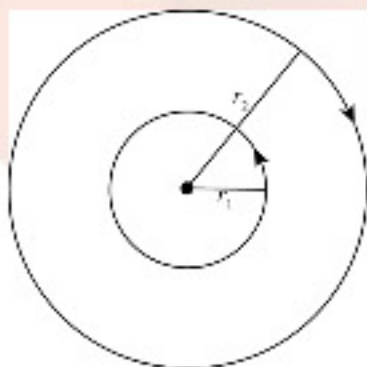
110. Given Data:

$$I_1 = 0.4 \text{ A}, I_2 = 0.6 \text{ A},$$

$$r_1 = 30 \times 10^{-2} \text{ m}$$

$$r_2 = 60 \times 10^{-2} \text{ m}$$

Consider the figure,



The expression for the magnetic field at O,

$$B_O = \frac{\mu_0}{4\pi} \cdot 2\pi \left[\frac{n_1 I_1}{r_1} - \frac{n_2 I_2}{r_2} \right]$$

Substitute the values in the above equation.

$$B_O = \frac{\mu_0}{4\pi} \cdot 2\pi \left[\frac{20 \times 0.4}{30 \times 10^{-2}} - \frac{20 \times 0.6}{60 \times 10^{-2}} \right]$$

Further simplification,

$$B_O = \frac{10}{3} \mu_0$$

111. The expression for the force per unit length is,

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi x}$$

The expression for the work done per unit length is,

$$W = F \cdot dx$$

$$W = \frac{\mu_0 I_1 I_2}{2\pi x} \cdot dx$$

$$W = \frac{\mu_0 I_1 I_2}{2\pi} \int \frac{dx}{x}$$

$$W = \frac{\mu_0 I_1 I_2}{2\pi} \log_e x$$

$$W \propto \log_e x$$

112. The correct matching is shown below,

(a)	High Retentivity	(iv)	Permanent magnet
(b)	High resistivity	(iii)	To decrease eddy current losses
(c)	Low coercivity	(i)	Telephone diaphragm
(d)	Negative susceptibility	(ii)	Diamagnet

Thus correct option is (2).

113. Due to the repulsion caused by emf induced, it is more difficult to push a magnet into a coil with a number of turns

The emf induced in the coil always opposes the motion of a magnet.

114. The power is expressed as,

$$P = I^2 R$$

Substitute the values in the given equation.

$$240 = 4^2 R$$

$$R = 15 \Omega$$

For L - R circuit,

$$V = IZ$$

$$V = I\sqrt{X_L^2 + R^2}$$

$$\frac{V^2}{I^2} = X_L^2 + R^2$$

Substitute the values in the above equation.

$$\frac{100^2}{4^2} = X_L^2 + 15^2$$

$$X_L^2 = 625 - 225$$

$$X_L^2 = 400$$

$$X_L = 20$$

And

$$X_L = L\omega$$

$$X_L = L(2\pi f)$$

$$20 = L(2\pi \times 50)$$

$$L = \frac{1}{5\pi} \text{ H}$$

115. The expression for the speed of the electromagnetic wave is,

$$c = \frac{E_0}{B_0}$$

$$3 \times 10^8 = \frac{60}{B_0}$$

$$B_0 = 2 \times 10^{-7} \text{ T}$$

116. From the de-Broglie wavelength of electron, the kinetic energy of the electron,

$$\lambda_e = \frac{h}{\sqrt{mK_e}}$$

$$K_e = \frac{h^2}{\lambda^2 \times 2m_e}$$

The kinetic energy of photon is,

$$K_p = hv = \frac{hc}{\lambda_p}$$

The required ratio is,

$$\frac{K_e}{K_p} = \frac{h^2 / \lambda^2 2m_e}{hc / \lambda}$$

$$\frac{K_e}{K_p} = \frac{h}{c\lambda \times 2m_e}$$

Substitute the given values in the above equation.

$$\frac{K_e}{K_p} = \frac{6.66 \times 10^{-34}}{3 \times 10^8 \times 1.2 \times 10^{-10} \times 2 \times 9.1 \times 10^{-31}}$$

$$\frac{K_e}{K_p} \approx \frac{1}{100}$$

117. Given Data,

$$r_1 = 5.5 \times 10^{-11} \text{ m}$$

$$v_1 = 4 \times 10^6 \text{ m/s}$$

The expression for the first excited state is,

$$r_n \propto \frac{n^2}{Z}$$

$$\frac{r_n}{2^2} = \frac{5.5 \times 10^{-11}}{1^2}$$

$$r_n = 22 \times 10^{-11} \text{ m/s}$$

The speed of the electron,

$$v_n \propto \frac{Z}{n}$$

$$v_n = \frac{4 \times 10^6}{2} = 2 \times 10^6 \text{ m/s}$$

The time period of revolution,

$$\begin{aligned}T &= \frac{2\pi e}{v_a} \\&= \frac{2 \times 3.14 \times 22 \times 10^{-11}}{2 \times 10^6} \\&= 6.908 \times 10^{-16} \text{ s}\end{aligned}$$

118. From the kinetic energy formula,

$$v^2 = \frac{2 \times KE}{m}$$

Substitute the given values in the above equation.

$$v^2 = \frac{2 \times 4 \times 1.6 \times 10^{-19}}{3.2 \times 10^{-21}}$$

$$v^2 = 400$$

$$v = 20 \text{ m/s}$$

The time required to travel a distance of 3.6 km is,

$$t = \frac{D}{v}$$

$$t = \frac{3.6 \times 10^3}{20}$$

$$= 180 \text{ s or } 3 \text{ min}$$

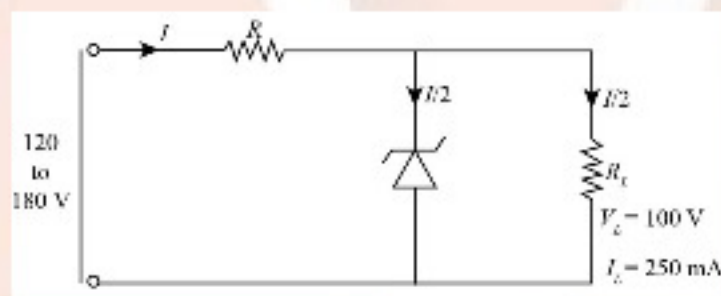
The number of particle decays in 3 min,

$$n = \frac{N_0 - N}{N_0} \times 100$$

$$n = \frac{N_0 - N_0 / 8}{N_0} \times 100$$

$$n = \frac{7}{8} \times 100 = 87.5\%$$

119. The figure below represents the electric circuit with zener diode.



$$R_L = \frac{V_L}{I_L}$$

$$= \frac{110}{250 \times 10^{-3}} = 440 \, \Omega$$

The current will equally shared by zener diode and resistance.

$$I = 2 \times 250 = 500 \, \text{mA}$$

Therefore,

$$\begin{aligned} R_s &= \frac{V_s}{I} \\ &= \frac{180 - 110}{0.5} \\ &= 140 \, \Omega \end{aligned}$$

120. The process of recovering the modulated signal from the modulated carrier wave is called demodulation.

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121. The energy is calculated as:

$$\begin{aligned}E_n &= -2.18 \times 10^{-18} \times \frac{Z^2}{n^2} \\&= -2.18 \times 10^{-18} \times \frac{4}{4} \\&= -2.18 \times 10^{-18} \text{ J}\end{aligned}$$

The radius is calculated as:

$$\begin{aligned}r_n &= 52.9 \times \frac{n^2}{Z} \\&= 52.9 \times \frac{4}{2} \\&= 105.8 \text{ pm}\end{aligned}$$

122. The energy is calculated as:

$$\begin{aligned}E &= h \frac{c}{\lambda} \\&= 6.6 \times 10^{-34} \text{ Js} \times \frac{3 \times 10^8 \text{ m/s}}{300 \times 10^{-9} \text{ m}} \\&= \frac{1.98 \times 10^{-25}}{300 \times 10^{-9} \text{ m}} \\&= 6.6 \times 10^{-19} \text{ J}\end{aligned}$$

The energy in eV is derived as:

$$1 \text{ eV} = 1.602 \times 10^{-19}$$

$$\begin{aligned}\text{Energy} &= \frac{6.6 \times 10^{-19}}{1.602 \times 10^{-19}} \\ &= 4.119 \text{ eV}\end{aligned}$$

The metals which are having less value of work function than the energy calculated, they can undergo photoelectric effect.

Therefore, metals are Mg, K and Na

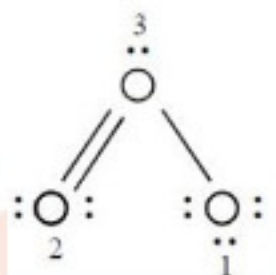
123. Since, we know that for isoelectronic species, radius is inversely proportional to effective nuclear charge.

Therefore, both Assertion (A) and Reason (R) are correct and (R) is correct explanation of (A).

124. The formal charge is calculated by using the formula shown below.

FC = Number of Valence electrons –

$$\left[\text{number of lone pair} + \frac{1}{2} \text{ number of bond pair} \right]$$

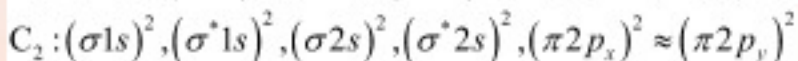
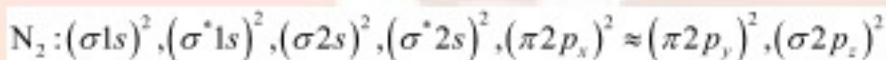


For first oxygen, formal charge is equal to $6 - [6 + 1] = -1$

For second oxygen, formal charge is equal to $6 - [4 + 2] = 0$

For third oxygen, formal charge is equal to $6 - [2 + 3] = 1$

125. The electronic configuration of the species which are having diamagnetic property (paired electrons) are shown below:



126. Total mass is $W_{\text{He}} + W_{\text{Ne}} = 4 \text{ g} \dots\dots\dots(\text{I})$

By using ideal gas equation;

$$PV = (n_{\text{He}} + n_{\text{Ne}})RT$$

$$PV = \left(\frac{W_{\text{He}}}{M_{\text{He}}} + \frac{W_{\text{Ne}}}{M_{\text{Ne}}} \right) RT$$

$$\frac{PV}{RT} = \left(\frac{W_{\text{He}}}{M_{\text{He}}} + \frac{W_{\text{Ne}}}{M_{\text{Ne}}} \right)$$

Substitute all the values.

$$\frac{1.23 \times 10}{0.082 \times 300} = \left(\frac{W_{\text{He}}}{4} + \frac{W_{\text{Ne}}}{20} \right)$$

$$50 = 25W_{\text{He}} + 5W_{\text{Ne}} \dots\dots (II)$$

By subtracting equations (II) from (I), we get

$$W_{\text{He}} = 1.5$$

$$W_{\text{Ne}} = 2.5$$

Mass percentage of neon is calculated as:

$$\text{Mass \% of Ne} = \frac{2.5 \times 100}{4}$$

$$= 62.5\%$$

127. The formula of first oxide is XO_2 which means one mole of X contains 2 mole of oxygen.

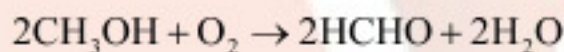
For second oxide, there is 40% of (X) which means there is 60% of oxygen.

Now, for 40% of X, we have $2 \times 60\% = 80\%$ of oxygen,
The moles of oxygen for 100% of X can be calculated as:

$$\begin{aligned}\text{Moles of oxygen} &= \frac{2 \times 60}{4} \\ &= 3 \text{ moles}\end{aligned}$$

Therefore, the formula of second oxide will be XO_3 .

128. The oxidation reaction of methanol to formaldehyde and water is shown below.



The standard enthalpy change for this reaction is calculated by using the expression shown below

$$\Delta H_{\text{reaction}}^{\circ} = \sum m \cdot \Delta H_f^{\circ}(\text{products}) - \sum n \cdot \Delta H_f^{\circ}(\text{reactants})$$

$$\begin{aligned}\Delta H_{\text{reaction}}^{\circ} &= \frac{1}{2} [2 \cdot \Delta H_f^{\circ}(\text{HCHO}) + 2 \cdot \Delta H_f^{\circ}(\text{H}_2\text{O})] - \\ &\quad \frac{1}{2} [2 \cdot \Delta H_f^{\circ}(\text{CH}_3\text{OH}) + \Delta H_f^{\circ}(\text{O}_2(\text{g}))]\end{aligned}$$

$$\begin{aligned}
 \Delta H_{\text{reaction}}^{\circ} &= \frac{1}{2} [2(-116) + 2(-286)] - \frac{1}{2} [2(-239) + (0)] \\
 &= \frac{1}{2} [-232 - 572] - \frac{1}{2} [-478] \\
 &= -163 \text{ kJ}
 \end{aligned}$$

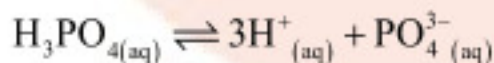
129. The expression of equilibrium constant for all the given reactions are shown below.

$$K_1 = \frac{[\text{H}^+][\text{H}_2\text{PO}_4^-]}{[\text{H}_3\text{PO}_4]}$$

$$K_2 = \frac{[\text{H}^+][\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}$$

$$K_3 = \frac{[\text{H}^+][\text{PO}_4^{3-}]}{[\text{HPO}_4^{2-}]}$$

Now, we have to calculate value of K for the following reaction.

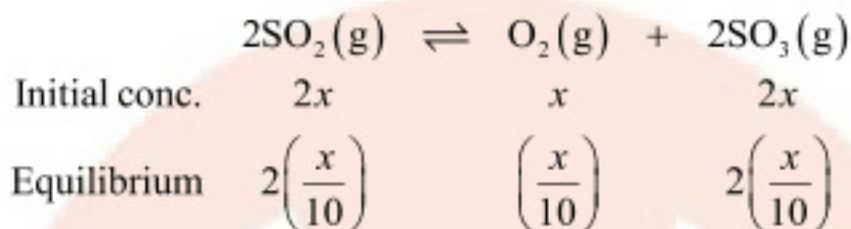


By multiplying K_1, K_2, K_3 , we get equilibrium constant of above reaction.

$$K_1 \times K_2 \times K_3 = K$$

130. Let's assume that number of moles of oxygen is 'x'

The dissociation reaction is as follows:



The equilibrium constant is calculated as:

$$K = \frac{\left(\frac{2x}{10}\right)^2}{\left(\frac{x}{10}\right)^2 \left(\frac{x}{10}\right)}$$

By simplifying the above equation, we get $x = 0.4$.

131. First calculate normality of H_2O_2 (N_2) as shown below.

$$\begin{aligned} \text{H}_2\text{O}_2 &= \frac{\text{volume of O}_2 \text{ at NTP}}{5.6} \\ &= \frac{20}{5.6} \\ &= 3.57 \end{aligned}$$

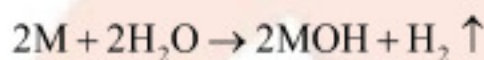
Apply the formula as follows:

$$N_1V_1 = N_2V_2$$

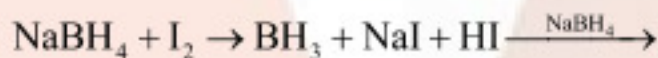
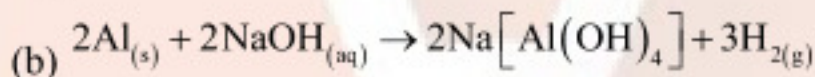
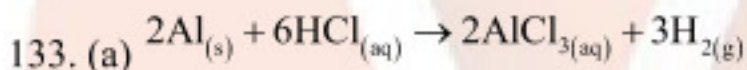
$$1 \times 500 = 3.57 \times V_2$$

$$V_2 = 140 \text{ mL}$$

132. When alkali metals reacts with water, they form metal hydroxide and hydrogen gas is liberated as follows:



Therefore, statement (1) is not correct.



Therefore, hydrogen is liberated in all the given reactions.

134. The species $[\text{SiF}_6]^{2-}$ exists but $[\text{SiCl}_6]^{2-}$ cannot exist because the size of chlorine atom is large. Therefore, $[\text{SiCl}_6]^{2-}$ is not known.

Thus, statement (2) is not correct.

135. Chloro-fluoro carbon releases chlorine atom which reacts with ozone and produce holes in the layer of ozone.

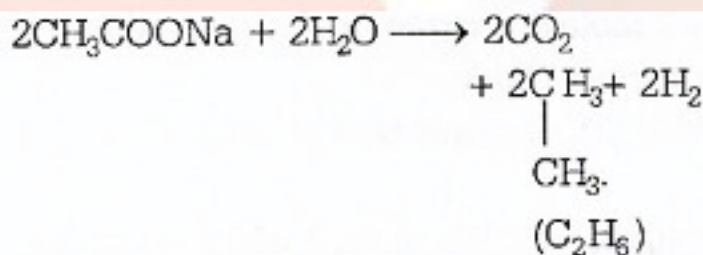
136. a) Electromeric effect is a temporary effect.

b) Hyper conjugation is a permanent effect.

c) For the separation of two liquids whose difference of boiling points is less, fractional distillation is used.

d) Different compounds are adsorbed on an adsorbent to different extents.

137. When an aqueous solution of sodium acetate is electrolysed, the reaction occurs as follows:



At anode (Y), C_2H_6 and CO_2 is formed.

At cathode (X), H_2 gas is liberated.

138. For showing the aromaticity,

The ring must possess $(4n + 2)\pi$ electrons.

The ring must be planar.

Therefore, statements (b) and (c) are correct.

139. For a body centered unit cell, the number of atoms is 2.

The density is derived by using the relation shown below.

$$d = \frac{zM}{N_A a^3}$$

$$\begin{aligned} d &= \frac{2 \times 24 \text{ g/mol}}{6 \times 10^{23} \text{ mol}^{-1} \times (400 \times 10^{-10} \text{ cm})^3} \\ &= 1.25 \text{ g/cm}^3 \end{aligned}$$

140. The change in freezing point is calculated as

$$\begin{aligned} \Delta T_f &= iK_f m \\ &= iK_f m \\ &= \frac{i \times K_f \times W_{\text{C}_2\text{H}_6\text{O}_2} \times 1000}{M_{\text{C}_2\text{H}_6\text{O}_2} \times W_{\text{H}_2\text{O}}} \\ &= \frac{1.86 \times 31 \times 1000}{62 \times 600} \\ &= 1.55 \text{ K} \end{aligned}$$

141. The osmotic pressure is calculated as:

$$\pi = iCRT$$

For 5.0 M urea,

$$\begin{aligned}\pi &= 1 \times 5 \times R \times 340 \\ &= 1700R\end{aligned}$$

For 1.5 M AB_3 type,

$$\begin{aligned}\pi &= 4.1 \times 1.5 \times R \times 300 \\ &= 1800R\end{aligned}$$

For 3 M AB type,

$$\begin{aligned}\pi &= 1.6 \times 3 \times R \times 300 \\ &= 1440R\end{aligned}$$

For 2.5 M AB_2 type,

$$\begin{aligned}\pi &= 2.5 \times 2.5 \times R \times 330 \\ &= 2062R\end{aligned}$$

Therefore, highest osmotic pressure is shown by the sample
2.5 M AB_2 type.

142. For a first order reaction, time can be calculated as:

$$t = \frac{2.303}{k} \log \frac{100 - 20}{100 - 80}$$

$$k = \frac{2.303}{90} \log \frac{80}{20}$$
$$= 0.015 \text{ min}^{-1}$$

The half-life is calculated as:

$$t_{1/2} = \frac{0.693}{k}$$
$$= \frac{0.693}{0.015}$$
$$= 45 \text{ min}$$

143. 1 mole of Ag^+ is equal to $\frac{1}{2}$ mole of Cu^{2+}

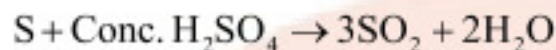
Therefore, for z moles of Ag^+ , we obtain $\frac{y}{2}$ mole of Cu^{2+}

Thus, $x = 2y$

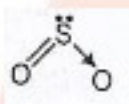
144. The difference of size in the gold particles causes the colloidal solution of gold to be in different colors like red, purple, blue and golden.

145. Copper matte is a mixture of Cu_2S and FeS

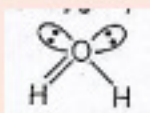
146. The reaction taking place is shown below.



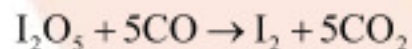
In sulphur dioxide, there is one lone pair present on central atom which is sulphur as follows:.



In water, there are two lone pairs present on central atom which is oxygen.

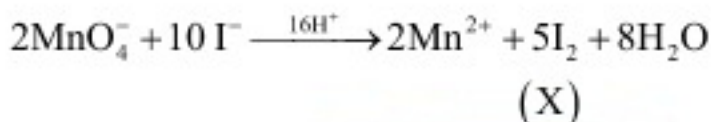


147. By using I_2O_5 , we can estimate carbon monoxide as follows:

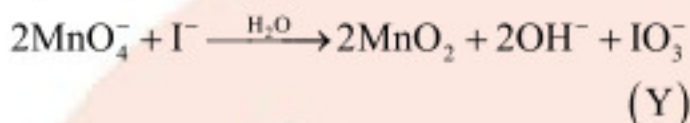


148. By taking xenon and fluorine in the ratio of 1:20 and by applying pressure of 70 bar at 573 K temperature, xenon hexafluoride is formed.

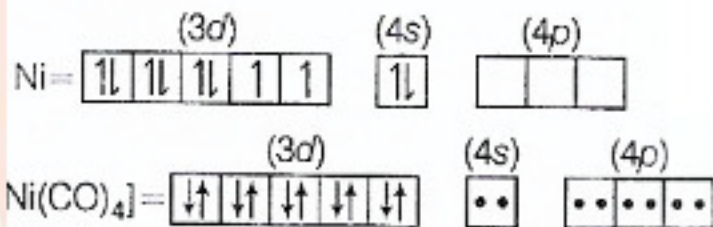
149. a) The product (X) formed is as follows:



b) The product (Y) formed is as follows:



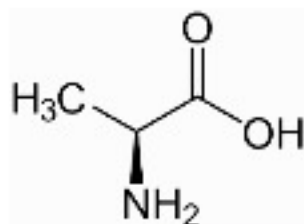
150. The tetrahedral complex is formed by $[\text{Ni}(\text{CO})_4]$ as CO is a strong ligand which causes pairing of unpaired electrons.



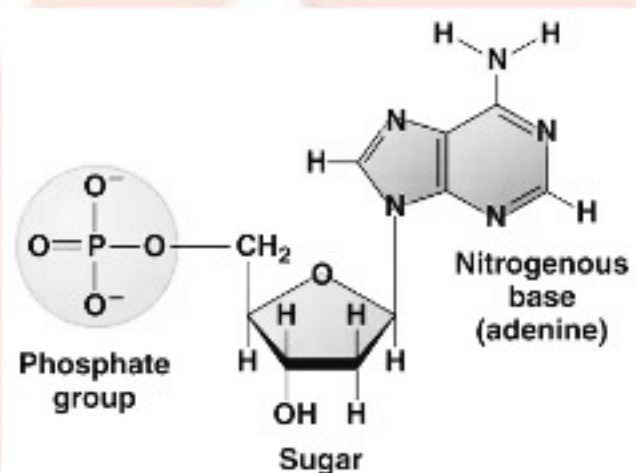
Hence, the complex is diamagnetic in nature.

151. $(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$ cannot be used as an initiator in ionic polymerization as it is a covalent compound and will not produce any free radical for chain initiation process.

152. b) The basic structure of alanine is

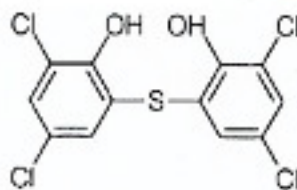


c) Nucleotide contains Base-Sugar and phosphate unit as shown below.

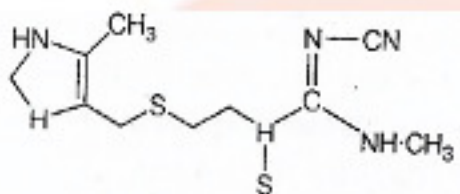


Therefore, statements (b) and (c) are incorrect.

153. The following structure is generally used to impart antiseptic properties in soap.

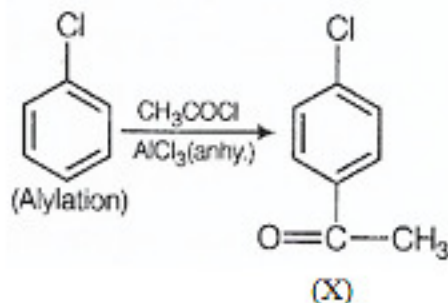


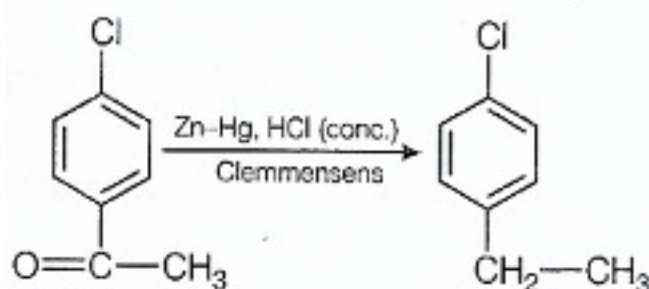
154. The drug which was discovered to avoid the interaction of histamine with the receptors present in the stomach wall is cimetidine. Its structure is shown below.



155. The species which is less bulkier is more reactive towards S_N2 reaction. Among the given species, CH_3X is least bulkier alkyl halide, therefore, its rate of reaction will be faster towards S_N2 reaction.

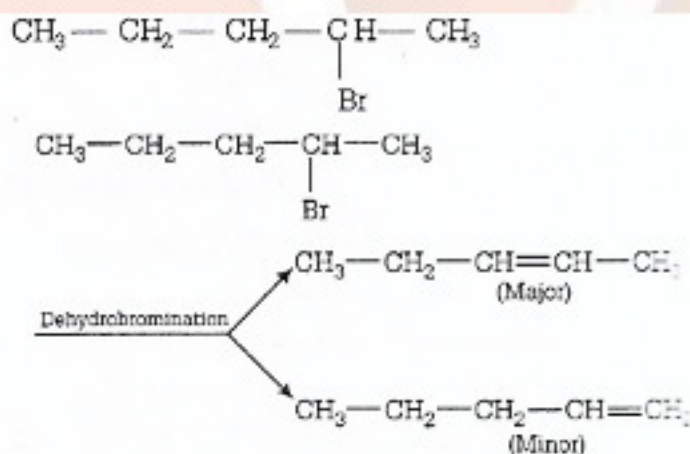
156. The product (Z) formed is shown below.





Therefore, Z is CH_2CH_3

157. Dehydrobromination of 2-bromopropane produce pent-2-ene as the major product.

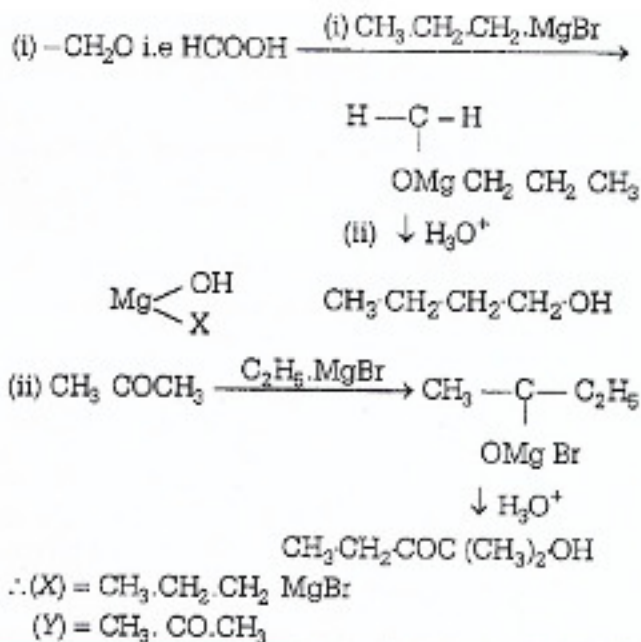


Therefore, statement (4) is not correct.

158. The increasing order of acidity is as follows:



159. The reaction is shown below.



160. The products (X) and (Y) formed are shown below.

